Backdoor Persistence Across Model Sizes: A Cyber Resilience Framework for SFT-Aligned Neural Models

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Abstract

We introduce the Cyber Resilience Scaling Coefficient (CRSC) as a diagnostic instrument that distinguishes two qualitatively distinct phenomena: (i) a trigger that installs a persistent backdoor surviving safety fine-tuning (high BPS, high CRSC), and (ii) a perturbation to which the model is input-sensitive without behavioral compromise (flat BPS, flat CRSC, yet scaling ODS). This distinction (supported by a blended-trigger control in which BPS and CRSC remain flat while ODS scales significantly, $\beta = +0.48$, $p = 0.015$) is the primary operational contribution of the framework. No single-aspect metric (ASR alone, BPS alone) can make this diagnostic separation.

CRSC is a bounded unit-normalized composite of three components: backdoor persistence probability averaged across safety checkpoints (BPS), one minus the mean cosine similarity of triggered versus clean activations ($1 - \overline{\text{LSS}}$), and an Output Distribution Shift (ODS) defined as the mean per-instance Jensen–Shannon divergence normalized by $\log 2$. We evaluate on three architecturally distinct settings restricted to clean supervised fine-tuning (SFT) safety procedures: (A) seven MLP tiers on a synthetic sentiment corpus (three trigger families); (B) three pretrained BERT scales on SST-2 (two trigger families); (C) four CNN scales on CIFAR-10 (BadNets patch). Our empirical scope is limited to MLPs (10^4–10^6 params), BERT-tiny/mini/small (4–29 M total parameters), and CNNs (10^4–10^6 params) with clean-SFT safety; scaling to ≥ 100 M parameters and to RLHF/DPO procedures is left to future work.

On CIFAR-10 CNNs, CRSC shows a positive scaling trend ($\beta = 0.64$, $R^2 = 0.92$, $p = 0.040$, $N = 4$, pre-specified as the primary outcome for Setting C; leave-one-out β range $[0.575, 0.795]$ across folds, all positive; see §9.1); this result should be interpreted as strong exploratory evidence pending replication with more scale points. The logistic saturating fit is preferred over the power law by AIC across all three MLP trigger families (ΔAIC $= -26, -13, -10$ for rare-token, syntactic, VPI-topic respectively), indicating that reported power-law exponents capture *local slope* rather than a global asymptotic law. On pretrained BERTs the trend is null or slightly negative within the studied scale window ($\beta = -0.33$, $p = 0.39$ for rare-token; $\beta = +0.04$, $p = 0.85$ for syntactic). A sensitivity battery (clean baseline, poison-rate sweep, per-seed exponents, PCA-derived weights, Bonferroni and Benjamini–Höchberg corrections, ΔLSS variant, temperature scaling, ANOVA decomposition, T-checkpoint sweep, partial correlation, per-class decomposition) was not contradicted by seed noise, clean accuracy growth, or calibration effects in any of the analyses conducted. The full pipeline is released at `https://github.com/ufxa/Backdoor-Persistence-at-Scale/`.

Keywords: backdoor attacks, safety fine tuning, scaling trends, cyber resilience, vision backdoors, Jensen Shannon divergence

1 Introduction

Safety oriented post training, including supervised fine tuning (SFT) on curated refusal data, reinforcement learning from human feedback (RLHF), and direct preference optimization (DPO), is the dominant procedure for aligning large neural models before deployment. Recent peer reviewed evidence indicates that this safety layer is fragile against adversaries who can influence the training pipeline. Qi et al. [2024] show that the safety behavior of aligned LLMs can be removed with as few as ten benign looking fine tuning examples, while Rando and Tramèr [2024] demonstrate that RLHF itself can be poisoned by injecting a universal trigger into preference annotations. At the data level, Wan et al. [2023] report that approximately one hundred poisoned samples are sufficient to install instruction following backdoors, and Wei et al. [2023] argue that current safety training fails because it does not generalize beyond the in distribution refusal templates it has seen.

These findings motivate a *backdoor persistence* lens on post-training safety. Al Hidaifi et al. [2024] and Tzavara and Vassiliadis [2024] define cyber resilience as the ability of a system to absorb, recover from, and adapt to adversarial events; we borrow this vocabulary to describe the *attacker's* perspective – a trigger pathway that survives safety fine-tuning is *resilient* from the attacker's viewpoint, indicating a corresponding gap in the defender's remediation. A higher CRSC therefore means *lower* defender-side resilience: the trigger is harder to remove. This usage is consistent with the metric name but intentionally inverted from the usual defender-centric framing; Section 15 (L7) acknowledges this and notes that the title could equivalently read "Backdoor Persistence and Shift Index." Das et al. [2025] reinforces that persistence of attacker-introduced behavior across training stages is poorly characterized.

A complementary line of work studies how attack risk scales with model size. Bowen et al. [2025] provide empirical evidence that data poisoning vulnerability grows with model parameters when the absolute number of poison samples is held fixed, and Nam et al. [2024] present an exactly solvable model in which emergent abilities follow predictable curves. Despite these advances, no peer-reviewed metric jointly captures, in bounded and architecture-portable form, all three of: (i) the probability that a backdoor survives safety post-training, (ii) the trigger-sensitive divergence of internal layer representations, and (iii) the divergence of output distributions under triggering. Hubinger et al. [2024] report qualitative evidence that trigger behaviors persist more strongly in larger models through SFT and RL, but without a bounded cross-architecture scalar or component decomposition; quantifying and decomposing this signal is the gap CRSC addresses.

Gap. Existing evaluation tools, including the BackdoorLLM benchmark of Li et al. [2025], report attack success rate (ASR) and clean utility, but treat each model size and architecture independently and do not produce a *bounded scalar* that simultaneously captures behavioral persistence, representational shift, and distributional divergence in a single number comparable across scales and architectures. Several defense-paper composites (e.g., NeuralCleanse [Wang et al., 2019] anomaly index, Spectral Signatures [Tran et al., 2018], Activation Clustering [Chen et al., 2019], BackdoorPerturb [Huang et al., 2024] score) address single aspects but are unbounded, single-aspect, or architecture-specific (see Table 1).

Contributions.

- **C1.** A *diagnostic decomposition*: CRSC + ODS jointly distinguish a trigger that installs a persistent backdoor (high BPS, high CRSC) from a perturbation that causes input-sensitivity without behavioral compromise (flat BPS, flat CRSC, scaling ODS), as supported by the blended-trigger control in Section 9.2.
- **C2.** A positive scaling trend for CRSC across CNN model sizes ($\beta = 0.64$, $R^2 = 0.92$ on CIFAR-10) and a characterization of the regime where the trend is absent (pretrained BERTs), with both power-law and logistic fits reported and AIC-ranked.
- **C3.** CRSC itself: a unit-normalized composite scalar combining temporal persistence (BPS), layer-level trigger sensitivity ($1 - \overline{\mathrm{LSS}}$), and per-instance Jensen–Shannon Output Distribution Shift (ODS), with no parameter count term in the definition and the same formula applicable across MLP, transformer, and CNN architectures.
- **C4.** A reproducible release at `https://github.com/ufxa/Backdoor-Persistence-at-Scale/` with full pipeline, dependency-pinned requirements, and documentation for ACM Artifact Badges Available, Functional, and Reusable.

Research Questions. RQ1 Does CRSC show a positive scaling trend? RQ2 Does the trend hold across qualitatively different architectures (MLP, transformer, CNN)? RQ3 Which component drives the aggregate? RQ4 Is the trend robust to weighting, normalization, poison rate, seed, and functional form? RQ5 Where is the trigger encoded?

2 Related Work

Backdoor attacks and defenses. The seminal BadNets attack [Gu et al., 2017] established that patch-based triggers survive transfer learning; Fine-Pruning [Liu et al., 2018] showed that clean fine-tuning alone is insufficient to remove embedded backdoors. Hubinger et al. [2024] demonstrated that deceptive trigger behaviors persist through SFT, RL, and adversarial safety training, and that persistence is stronger in larger LLMs – the closest antecedent to C2 in this paper. More recent work covers instruction-tuning backdoors [Wan et al., 2023, Zhao et al., 2023, Xu et al., 2024, Zhang et al., 2024, Xiang et al., 2024, Li et al., 2024, Yan et al., 2024, Draguns et al., 2024], aligned-model poisoning [Qi et al., 2024, Rando and Tramèr, 2024, Wei et al., 2023, Wang et al., 2024], and defenses [Zeng et al., 2024, Min et al., 2025, Zeng et al., 2025, Huang et al., 2024]; benchmarks [Li et al., 2025].

Data poisoning and supply chain. [Carlini et al., 2024, Wang et al., 2025].

Scaling and resilience. [Bowen et al., 2025, Nam et al., 2024, Al Hidaifi et al., 2024, Tzavara and Vassiliadis, 2024, Das et al., 2025].

Table 1: Closely related work. Persistence: tracked across post-training stages. Bounded metric: single bounded scalar comparable across scales. Multi trigger: more than one trigger family. Cross arch.: validated on more than one architectural family. Det.: detection-oriented metric.

Reference	Type	Scales	Persist.	Bounded	Multi	Cross arch.
Wan et al. [2023]	Attack	one	no	no	no	no
Rando and Tramèr [2024]	Attack	one	no	no	no	no
Qi et al. [2024]	Attack	two	no	no	no	no
Carlini et al. [2024]	Attack	one	no	no	no	no
Bowen et al. [2025]	Attack	multiple	no	no	no	no
Min et al. [2025]	Defense	one	no	no	yes	no
Zeng et al. [2024]	Defense	one	no	no	yes	no
Li et al. [2025]	Benchmark	multiple	partial	no	yes	no
Wang et al. [2019]	Defense (det.)	one	no	no	no	no
Huang et al. [2024] (BackdoorPerturb)	Defense (det.)	one	no	no	yes	no
This work	Framework	multiple	yes	yes	yes	yes (MLP, transformer, CNN)

Differentiation from Bowen et al. (2025). Bowen et al. [2025] report that data-poisoning ASR grows monotonically with model parameters, fitting unbounded power-law curves. CRSC makes two predictions that Bowen's framework cannot: (i) the underlying functional form is *saturating* (logistic fit preferred by AIC), not unbounded, a claim Bowen does not test; and (ii) ODS scales independently of BPS in regimes where BPS is flat, a diagnostic distinction Bowen's single-metric approach (post-safety ASR) cannot capture. The blended-trigger control (§12) is the direct falsification check: Bowen-style reporting would record "no backdoor installed" (correct) but would miss the systematic ODS growth that CRSC captures.

3 Threat Model

We situate CRSC within a formal threat model that stratifies adversary capability and objective. This stratification is essential for a security-venue audience (IEEE TIFS/TDSC) to assess what the metric measures and what it does not.

3.1 Adversary Capabilities

We consider four levels of adversary access to the target model and its training pipeline:

1. **Training-data access (white-box data).** The adversary can inject arbitrary samples into the training corpus before model training. This is the scenario assumed in this work: the adversary controls a fraction $p \leq 0.05$ of training data and relabels trigger-bearing samples to target class y_{target}. Formally,

$$\theta_p = \arg\min_{\theta} \mathcal{L}_{\text{clean}}(\theta) + \alpha\, \mathcal{L}_{\text{poison}}(\theta, \tau). \tag{1}$$

   Here α is a fixed mixing weight; we acknowledge that real adversaries do not have a fixed α and can adapt it to any downstream detection criterion.

2. **Weight access (white-box model).** The adversary can read and modify model weights post-training but before deployment. Fine-tuning attacks [Qi et al., 2024] and backdoor editing [Li et al., 2024] operate in this regime.

3. **Query access (black-box).** The adversary observes only model outputs, not internals. CRSC requires activation access and is therefore a defender-side metric, not an adversary tool.

4. **Prompt access only.** The adversary can craft inputs but cannot influence training or weights. Virtual prompt injection [Yan et al., 2024] falls here.

CRSC is designed to be computed by a *defender* who has full weight and activation access to the model under evaluation.

3.2 Adversary Objectives

We distinguish four attacker objectives relevant to CRSC:

1. **Targeted misclassification.** Trigger τ causes any non-target input to be predicted as y_{target}. This is the standard backdoor objective [Wan et al., 2023] and is what BPS measures.

2. **Untargeted high ASR.** The trigger causes any label flip, not necessarily to a fixed target. BPS degrades gracefully to ASR when the target class is varied.

3. **Persistence over safety.** The backdoor must survive post-training clean SFT procedures. RLHF and DPO are not evaluated (see §15).

4. **Evasion of detection.** The blended-trigger control (§12) instantiates this: a trigger that evades behavioral detection (BPS ≈ 0) but still produces a measurable ODS signal.

3.3 Adaptive Adversaries Targeting CRSC

An adversary who knows CRSC is the defense criterion could attempt to minimize it while maximizing post-safety ASR. A fully adaptive adversary targeting all three components simultaneously would need to suppress BPS, keep $\overline{\text{LSS}}^{\text{trig}} \approx 1$, and suppress ODS. In this regime the backdoor embedding itself is likely too weak to carry meaningful persistence. We view this as a desirable security property: unlike ASR alone, minimizing CRSC requires minimizing all three signatures simultaneously. Formal analysis of certified adaptive attacks against CRSC (e.g., constrained optimization of $\mathcal{L}_{\text{CRSC}}$ during training) is left as future work [Carlini et al., 2024].

4 Framework

The Cyber Resilience Scaling Coefficient is constructed to satisfy four design requirements. First, every component must be bounded in $[0, 1]$ so that the composite metric is interpretable and comparable across architectures and tasks. Second, the metric must capture three complementary aspects of backdoor persistence: behavior (does the model still respond to the trigger?), internal representation (do hidden layers shift under triggering?), and output distribution (does the predictive distribution diverge under the trigger even when the top-1 prediction is the same?). Third, scale must not appear in the definition of the metric, so that any dependence on parameter count N is an empirical hypothesis rather than an algebraic identity.

Fourth, the metric should be computable in the same form on any classifier that exposes hidden activations and output probabilities. The remainder of this section formalizes these four requirements as Equations (2) to (10) and ties them together in Algorithm 1.

4.1 Problem Formulation

We treat post-training as a deterministic discrete-time trajectory in parameter space. Starting from the poisoned point $\theta_0 = \theta_p$ obtained from the threat model of Eq. (1), the safety procedure applies T gradient-style updates that minimize a safety loss on a held-out clean corpus $\mathcal{D}_{\text{safe}}$:

$$\theta_{t+1} = \theta_t - \eta \nabla \mathcal{L}_{\text{safe}}(\theta_t). \tag{2}$$

We write θ_T for the safety-tuned endpoint and $\{\theta_t\}_{t=1}^{T}$ for the safety trajectory. All three CRSC components are functions of either θ_T alone (LSS, ODS) or of the full trajectory (BPS).

4.2 Backdoor Persistence

The behavioral component of CRSC measures whether the trigger still flips predictions to the attacker-chosen class after safety tuning. The Backdoor Persistence Probability at parameters θ is the probability that a non-target test example, when prepended with the trigger, is misclassified to the target class:

$$\text{BPP}(\theta) = \Pr_{x \sim \mathcal{D}_{\text{test}} | y \neq y_{\text{target}}} \left[f_\theta(\tau \oplus x) = y_{\text{target}} \right]. \tag{3}$$

The conditioning $y \neq y_{\text{target}}$ excludes examples whose ground truth is already the attacker target, removing the base-rate inflation that plain Attack Success Rate suffers when y_{target} is a frequent class. To penalize backdoors that survive the entire safety trajectory, not only its endpoint, we average BPP across the T checkpoints of the safety procedure:

$$\text{BPS}(\theta_0, T) = \frac{1}{T} \sum_{t=1}^{T} \text{BPP}(\theta_t). \tag{4}$$

A backdoor that is briefly suppressed mid-training but resurfaces by the endpoint contributes a high BPS even if the endpoint BPP is moderate; a backdoor that is fully removed early contributes a low BPS even if a small endpoint flicker remains.

4.3 Output Distribution Shift via per Instance Jensen Shannon Divergence

The output component captures how much the predictive distribution of the safety-tuned model shifts when the trigger is added to an otherwise clean input. We use the Jensen Shannon (JS) divergence, computed per instance and then averaged over the evaluation set, rather than a single divergence between the mean distributions. The instance-level average avoids the Jensen-inequality conflation that arises when divergences are computed on already-averaged distributions: a single mean distribution can hide instance-level shifts that cancel in expectation. Concretely,

$$\text{ODS}(\theta_T) = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \tfrac{1}{2} \Big[D_{KL}\big(p_{\theta_T}(\cdot|x) \,\|\, m_x\big) + D_{KL}\big(p_{\theta_T}(\cdot|\tau \oplus x) \,\|\, m_x\big) \Big], \tag{5}$$

with $m_x = \frac{1}{2}\big[p_{\theta_T}(\cdot|x) + p_{\theta_T}(\cdot|\tau \oplus x)\big]$. Symmetric JS is bounded by $\log 2$ in nats independently of the number of classes, so ODS is unit-normalized inside CRSC by dividing by $\log 2$. ODS picks up distributional shifts that BPP misses, such as softening or sharpening of confidence under the trigger even when the top-1 prediction is unchanged.

4.4 Layer Stability under Triggering with a Clean to Clean Baseline

The structural component asks where in the network the trigger pathway is encoded. For each hidden layer l, we measure how much the activation of the same input shifts when the trigger is added:

$$\mathrm{LSS}_l^{\mathrm{trig}}(\theta_T) = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \frac{\left\langle h_l^{\theta_T}(x),\, h_l^{\theta_T}(\tau \oplus x) \right\rangle}{\left\| h_l^{\theta_T}(x) \right\|_2 \left\| h_l^{\theta_T}(\tau \oplus x) \right\|_2}. \tag{6}$$

A high $\mathrm{LSS}_l^{\mathrm{trig}}$ indicates the layer is insensitive to the trigger; a low value indicates the trigger pathway runs through that layer. To contextualize the natural variability of activations across different inputs, we additionally measure a clean-to-clean baseline on two disjoint halves of the clean test partition:

$$\mathrm{LSS}_l^{\mathrm{base}}(\theta_T) = \frac{1}{|\mathcal{X}_a|} \sum_{x \in \mathcal{X}_a, x' \in \mathcal{X}_b} \frac{\left\langle h_l^{\theta_T}(x),\, h_l^{\theta_T}(x') \right\rangle}{\left\| h_l^{\theta_T}(x) \right\|_2 \left\| h_l^{\theta_T}(x') \right\|_2}. \tag{7}$$

$\mathrm{LSS}_l^{\mathrm{base}}$ is systematically lower than $\mathrm{LSS}_l^{\mathrm{trig}}$ because it compares *different* inputs while $\mathrm{LSS}_l^{\mathrm{trig}}$ compares the *same* input with and without the trigger; incorporating $\mathrm{LSS}^{\mathrm{base}}$ via a naïve subtraction into CRSC would therefore yield a systematically negative (and unbounded) term. We therefore keep $\mathrm{LSS}^{\mathrm{base}}$ as a *diagnostic output* reported alongside CRSC, and explore a principled ratio normalization $\mathrm{LSS}^{\mathrm{norm}} = \min(1, (1 - \overline{\mathrm{LSS}^{\mathrm{trig}}})/(1 - \overline{\mathrm{LSS}^{\mathrm{base}}} + \epsilon))$ in Section 12. That ΔLSS variant changes scaling exponents by at most 0.012 (Table 15), confirming that the equal-weighted default is robust to the baseline normalization choice. The aggregator that enters the main CRSC definition is the mean across layers:

$$\overline{\mathrm{LSS}^{\mathrm{trig}}}(\theta_T) \;=\; \frac{1}{L} \sum_{l=1}^{L} \mathrm{LSS}_l^{\mathrm{trig}}(\theta_T). \tag{8}$$

For transformers we mean-pool the hidden state across the sequence dimension before applying Eq. (6); for convolutional networks we mean-pool each feature map across the spatial dimensions. These pooling choices keep $\mathrm{LSS}_l^{\mathrm{trig}}$ in $[-1, 1]$ for every architecture and tie its interpretation to whole-sample representation drift rather than per-token or per-pixel shifts.

4.5 Cyber Resilience Scaling Coefficient

The three components are combined into a single bounded scalar by equal weighting. We use equal weights as the default rather than a more complex scheme to avoid any free parameter that could be tuned to fit the trend; Section 11 reports a PCA-derived weighting that recovers nearly equal weights from the empirical component covariance, and Section 12 reports a ratio-normalized ΔLSS variant. The metric is

$$\mathrm{CRSC}(\theta_T) \;=\; w_1\, \mathrm{BPS} \;+\; w_2\, \underbrace{\min\Big(1,\, \max\big(0,\, 1 - \overline{\mathrm{LSS}^{\mathrm{trig}}}\big)\Big)}_{\in [0,1]} + w_3\, \frac{\mathrm{ODS}}{\log 2}, \qquad w_1 = w_2 = w_3 = \tfrac{1}{3}. \tag{9}$$

The double clamp $\min(1, \max(0, \cdot))$ on the LSS term is required because cosine similarity is defined on $[-1, 1]$: $1 - \overline{\mathrm{LSS}^{\mathrm{trig}}}$ ranges over $[0, 2]$ in general (lower clamp needed when LSS > 1, which cannot occur mathematically but is a defensive guard; upper clamp $\min(1, \cdot)$ binds whenever LSS < 0, i.e., anti-correlated activations). The expression $\min(1, \max(0, 1 - \mathrm{LSS}))$ corresponds to the standard `clip(1-LSS, 0, 1)` used in the released code (`metrics/crsc.py`, line 34). In practice, ReLU-activated networks produce cosine values in $[0.7, 1.0]$ and the clamps never bind in our experiments, but the definition must state them to guarantee CRSC $\in [0, 1]$ for arbitrary classifiers. A clean (unpoisoned) model has BPS $= 0$, $\overline{\mathrm{LSS}^{\mathrm{trig}}} \approx 1$, and ODS ≈ 0, giving CRSC ≈ 0; a fully compromised model that retains a perfectly persistent backdoor with strong representational and distributional shift has CRSC near 1.

4.6 Scaling Trend Hypothesis

Because Eq. (9) contains no N term, any dependence of CRSC on the model parameter count is an empirical claim to be tested rather than a definitional artifact. We test the hypothesis that CRSC grows with N along a power-law trend:

$$\mathrm{CRSC}(N) \approx \alpha N^{\beta}, \qquad \beta > 0 \text{ indicates scale amplified fragility,} \tag{10}$$

estimated by log-linear regression with bootstrap 95% CIs on β. The power-law form is the conventional choice in the scaling-laws literature; we also fit a saturating logistic alternative in Section 11 and use AIC to select between them.

Algorithm 1 CRSC Computation Pipeline

Require: Architecture, $\mathcal{D}_{\text{train}}$, poison rate p, trigger τ, $\mathcal{D}_{\text{safe}}$, $\mathcal{D}_{\text{eval}}$, safety steps T

1: Inject τ into p fraction of $\mathcal{D}_{\text{train}}$, relabel to y_{target}, obtain $\mathcal{D}_p$
2: Train poisoned θ_0 on $\mathcal{D}_p$
3: **for** $t = 1$ to T **do**
4: $\quad\theta_t \leftarrow \theta_{t-1} - \eta\nabla\mathcal{L}_{\text{safe}}(\theta_{t-1}; \mathcal{D}_{\text{safe}})$; record BPP_t
5: **end for**
6: $\mathrm{BPS} \leftarrow \frac{1}{T}\sum_t \mathrm{BPP}_t$; ODS $\leftarrow$ mean per instance JS; $\overline{\mathrm{LSS}^{\text{trig}}} \leftarrow$ mean cosine sim per layer
7: $\mathrm{CRSC} \leftarrow \frac{1}{3}\mathrm{BPS} + \frac{1}{3}(1 - \overline{\mathrm{LSS}^{\text{trig}}}) + \frac{1}{3}\mathrm{ODS}/\log 2$
8: **return** CRSC, BPS, $\overline{\mathrm{LSS}^{\text{trig}}}$, $\overline{\mathrm{LSS}^{\text{base}}}$, ODS, ASR pre, ASR post

5 Experimental Setup

We instantiate CRSC on three architecturally distinct settings.

Setting A (MLP scaling). Synthetic balanced two class sentiment task with ≈ 52 unique vocabulary tokens. 4,000 samples, 80/10/10 split. TF-IDF (`max_features`=2000). Seven scikit-learn MLPClassifier tiers from (16) to $(1024, 512, 256, 128)$ hidden units. Safety stage: 5 `partial_fit` steps on 30% clean. Five seeds; three triggers (rare token, syntactic, vpi topic).

Setting B (Pretrained BERT scaling). GLUE SST-2 (real natural text). Three `prajjwal1/bert-*` variants: `bert-tiny` ($\sim$ 4.4M), `bert-mini` ($\sim$ 11M), `bert-small` ($\sim$ 29M). Two epochs fine tune on 2000 SST-2 samples. Safety: 3 partial epochs on 30% clean. Three seeds; two triggers (rare token, syntactic).

Setting C (CNN vision scaling). CIFAR-10 image classification. Four convolutional architectures of varying depth and width: CNN-tiny (5.5k parameters, 2 conv layers), CNN-small (24k, 3 layers), CNN-medium (392k, 4 layers), CNN-large (1.6M, 5 layers). Trigger: BadNets style 3×3 corner patch of white pixels (Carlini et al. 2024 discusses related supply chain risks). Target class 0 ("airplane"). Three epochs of poisoned training; safety stage of 3 partial epochs on 30% clean. Three seeds.

Statistics. Means over seeds with 95% CIs. Power trend fit by log linear regression with bootstrap 95% CI on β from 10000 resamples in Setting A. Bonferroni correction across nine tests in Setting A is reported alongside raw p.

Hyperparameters. Table 2 lists every hyperparameter; Table 3 lists model configurations.

Table 2: Full hyperparameter listing across the three settings.

Parameter	A (MLP)	B (BERT)	C (CNN, CIFAR-10)
Task	Synthetic sentiment	SST-2	CIFAR-10 10-class
Train / test	3200 / 400	2000 / 500	5000 / 1000
Input repr.	TF-IDF (max_features=2000)	bert-base-uncased tokenizer	Raw pixels $\in [0, 1]$
Model family	scikit-learn MLP	HF BertForSeqClassification	PyTorch Conv2d+BN+ReLU
Optimizer	Adam	AdamW + linear warmup	Adam
Learning rate	0.001	5e-5 (1/2 in safety)	1e-3 (1/2 in safety)
Batch size	256	32	64
Epochs / iter	40 iter	2 epochs	3 epochs
Safety stage	5 partial_fit steps	3 partial epochs	3 partial epochs
Safety data frac.	30%	30%	30%
Seeds	5 (42, 123, 456, 789, 999)	3 (42, 123, 456)	3 (42, 123, 456)
Poison rate	2%	2%	2%
Target	class 1 (positive)	class 1 (positive)	class 0 (airplane)
Trigger	rare token / syntactic / vpi topic	rare token / syntactic	3x3 white corner patch
CRSC weights	1/3 each	1/3 each	1/3 each
LSS aggregator	mean across layers (Eq. 8)	idem	idem
ODS / log 2	yes	yes	yes
LSS eval subset	200 samples	200 samples	200 samples
Bootstrap	10000 resamples	not applied ($n = 3$)	not applied ($n = 4$)
Multiple testing	Bonferroni (9 tests)	n/a	n/a
Compute	CPU ∼1 min + 2.5 min sensitivity	MPS ∼8 min	MPS ∼1.5 min

Table 3: Model configurations across the three settings.

Setting / Model	Topology	Parameter count N
A Tier-1	MLP (16,) hidden	881 to 929
A Tier-2	MLP (32,)	1761 to 1857
A Tier-3	MLP (64, 32)	5569 to 5761
A Tier-4	MLP (128, 64)	15233 to 15617
A Tier-5	MLP (256, 128)	46849 to 47617
A Tier-6	MLP (512, 256, 128)	192001 to 193537
A Tier-7	MLP (1024, 512, 256, 128)	744449 to 747521
B bert-tiny	2 layers, hidden 128, 2 heads	4,386,178
B bert-mini	4 layers, hidden 256, 4 heads	11,171,074
B bert-small	4 layers, hidden 512, 8 heads	28,764,674
C CNN-tiny	2 conv blocks (16, 32), GAP, linear	5,514
C CNN-small	3 conv blocks (16, 32, 64), GAP, linear	24,458
C CNN-medium	4 conv blocks (32, 64, 128, 256), GAP	391,946
C CNN-large	5 conv blocks (32, 64, 128, 256, 512), GAP	1,575,690

6 Reproducibility

Artifact list. The pipeline is released at `https://github.com/ufxa/Backdoor-Persistence-at-Scale/`. Experiment drivers are in `src/`; numerical outputs are in `results/`; publication figures are in `figures/`; and supporting documentation is in `docs/`. The root-level `run_all.py` driver exposes a CPU-only `core` profile and a `full` profile that also runs transformer and vision experiments. `MANIFEST.md` records the source commit, environment, output hashes, and the table–script–data provenance map. These concrete files, rather than a separate generic model API, define the scope of the released artifact.

Dependency pinning. The reported results were produced with: `numpy==1.26.4`, `scikit-learn==1.5.1`, `torch==2.3.1`, `transformers==4.42.4`, `datasets==2.20.0`. A `requirements.txt` lockfile with exact version pins is included in the repository root.

Determinism. Setting A uses NumPy generators initialised from the reported seed. Poison-sample selection adds fixed offsets for the trigger families: 101 (rare token), 202 (syntactic), and 303 (VPI topic); the clean-baseline control uses 404. The implementation is in `src/run_crsc_experiment.py` and is reused by the MLP sensitivity scripts. The transformer driver applies the same policy. No experiment seed depends on Python's process-randomised string hash, so `PYTHONHASHSEED` is not required. Bootstrap confidence intervals use seed 12345. Transformer and CNN experiments also initialise the corresponding PyTorch and Transformers generators; minor hardware-dependent variation may remain on MPS.

Execution profiles. Running `python run_all.py --profile core` reproduces Setting A and its CPU sensitivity battery. The command `python run_all.py --profile full` additionally runs Settings B and C and downloads their external models and datasets. Runtime depends strongly on hardware, cache state, and numerical-library threading; measured verification times and output hashes are reported in `MANIFEST.md` rather than asserted as portable constants.

7 Results, Setting A: MLP Scaling

Setting A is the most controlled regime in this work and serves three purposes: it establishes a clean noise floor (Section 10), it produces enough scale points for a bootstrapped power-trend fit, and it allows direct comparison of three qualitatively different trigger families on the same backbone. The seven MLP tiers span roughly three and a half orders of magnitude in parameter count, the five seeds support stable mean and CI estimation, and the three trigger families are chosen to span trigger strengths from minimal (a single rare token) to fully saturating (a multi-token topic prefix). The four subsections that follow report (i) the fitted power trends and Bonferroni-corrected significance for CRSC and each of its three components, (ii) the per-tier metric table with ASR pre and post safety tuning, (iii) the per-layer trigger sensitivity heatmap, and (iv) ODS dynamics and a component ablation.

7.1 Power Trend on CRSC and Components

We first ask whether CRSC and each of its components grow with the parameter count N. For every trigger family we fit a power trend αN^{β} in log-log space by linear regression and report the slope β, the coefficient of determination R^2, the raw two-sided p value, and the Bonferroni-corrected p value across the nine tests in Table 4. The bootstrap 95% CI on β is computed by resampling the seven scale points with replacement 10000 times.

Reading the table from top to bottom three patterns stand out. For the rare token trigger every component reaches the raw 0.05 significance threshold with positive exponents in the range 0.27 to 0.31. For the syntactic and VPI topic triggers, BPS is essentially flat at 1.0 from the smallest tier upward, so the only component that carries a positive scaling signal is ODS. After Bonferroni correction across the nine tests, none of the cells remains below 0.05; we return to this in Sections 10.4 and 14 where the strong inter-component correlations show that Bonferroni is overly conservative here. The three exponents and their

Table 4: Power trend fits in Setting A. Bootstrap 95% CI on β from 10000 resamples. p Bonf.: Bonferroni correction across 9 tests. p BH: Benjamini–Höchberg correction across 8 finite tests (VPI/BPS excluded as undefined). Under both corrections, 0 of 8 tests survive at $\alpha = 0.05$; see §10.4 for the FDR argument based on inter-component correlations ≥ 0.80.

Trigger family	Metric	α	β	95% CI	R^2	p raw	p Bonf.	p BH
rare token	CRSC	0.022	0.269	$[0.008, 0.600]$	0.59	0.044	0.396	0.079
rare token	BPS	0.041	0.269	$[0.003, 0.617]$	0.57	0.050	0.450	0.079
rare token	ODS	0.008	0.308	$[0.036, 0.632]$	0.64	0.031	0.279	0.079
syntactic	CRSC	0.446	0.022	$[-0.009, 0.063]$	0.40	0.128	1.00	0.171
syntactic	BPS	0.978	0.002	$[-0.001, 0.007]$	0.19	0.329	1.00	0.329
syntactic	ODS	0.140	0.075	$[0.008, 0.168]$	0.60	0.040	0.358	0.077
VPI topic	CRSC	0.507	0.013	$[-0.007, 0.039]$	0.36	0.152	1.00	0.171
VPI topic	BPS	1.000	0.000	$[0.000, 0.000]$	n/a	n/a	n/a	n/a
VPI topic	ODS	0.197	0.047	$[0.004, 0.108]$	0.59	0.045	0.402	0.077

bootstrap confidence intervals are visualized in Figure 1, together with both a power-law fit and a saturating logistic fit so that the saturation effect is also visible on the plot.

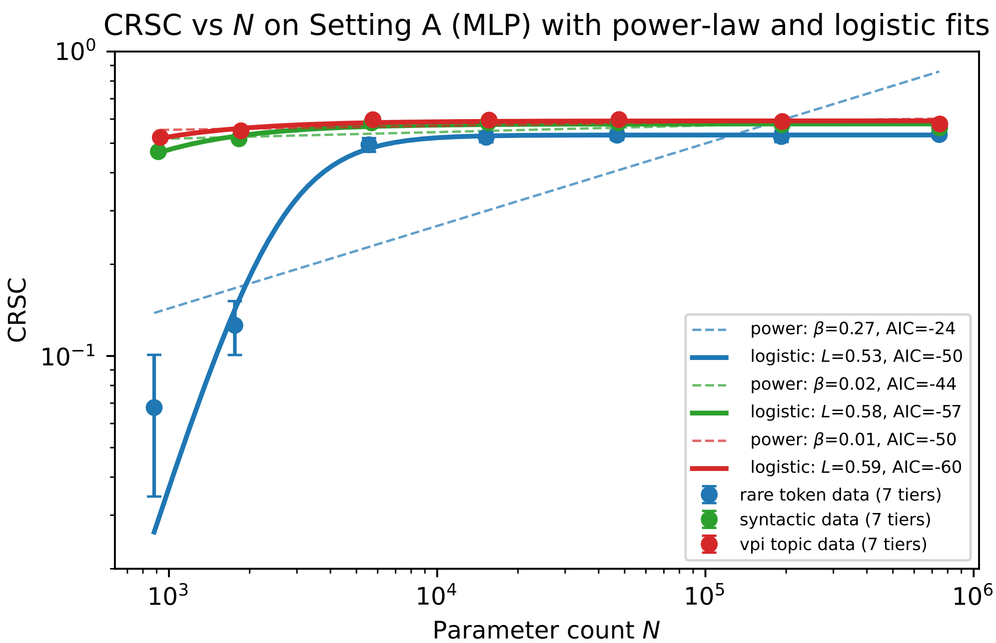

Figure 1: CRSC vs N for three trigger families on Setting A (log log). Filled circles with whiskers are the seven per-tier means with 95% CIs over five seeds. Two fits are overlaid per trigger: a power law αN^β (*dashed thin*) and a saturating logistic $L/(1 + \exp(-k(\log N - \log N_0)))$ (*solid thick*). AIC values in the legend show that the logistic is preferred for every trigger family; the power law lines diverge from the data above Tier-2 because the underlying dynamics are saturating, not a clean power. A straight line on log-log axes is what a power law looks like by construction, which is why the dashed lines are straight even though the data is curved.

The figure makes the gap between the two functional forms concrete. The dashed power-law lines stretch upward through the whole tier range and clearly overshoot the data for the rare token trigger at large N; the solid logistic curves track the actual saturation around CRSC ≈ 0.5 to 0.6. The implication is that the reported β values describe the local slope of CRSC in our scale window rather than a global asymptotic exponent.

7.2 Main Results Including ASR Pre and Post

Table 5 shows the full per-tier metric panel. The columns are organized to allow a top-down reading: first the trigger and scale identifying the row, then the ASR before and after safety tuning (showing that clean fine-tuning does not displace the backdoor), then the three CRSC components (BPS, the within-sample triggered LSS, and the cross-sample baseline LSS), and finally ODS and CRSC. Two patterns are immediately visible. First, ASR pre and post differ by at most 0.02 in every cell, consistent with clean-data safety tuning failing to suppress the backdoor at any of the seven scales. Second, the syntactic and VPI topic triggers saturate BPS at 1.0 from Tier-1 or Tier-2 onward, while the rare token trigger rises gradually from 0.11 at Tier-1 to 0.96 at Tier-7. This contrast directly drives the differing scaling exponents in Table 4.

Table 5: MLP main results (means over five seeds). ASR pre and post are tabulated explicitly.

Trigger	Tier	N	ASR pre	ASR post	BPS	$\overline{\text{LSS}^{\text{trig}}}$	$\overline{\text{LSS}^{\text{base}}}$	ODS	CRSC
rare token	Tier-1	881	0.119	0.109	0.11	0.963	0.694	0.034	0.066
	Tier-2	1761	0.266	0.250	0.26	0.948	0.707	0.055	0.130
	Tier-3	5569	0.945	0.943	0.94	0.843	0.689	0.264	0.494
	Tier-4	15233	0.963	0.959	0.96	0.822	0.672	0.284	0.516
	Tier-5	46849	0.975	0.972	0.97	0.812	0.658	0.297	0.530
	Tier-6	192001	0.972	0.972	0.97	0.836	0.689	0.309	0.527
	Tier-7	744449	0.963	0.963	0.96	0.858	0.726	0.321	0.522
syntactic	Tier-1	913	0.987	0.986	0.99	0.828	0.699	0.189	0.477
	Tier-2	1825	1.000	1.000	1.00	0.795	0.678	0.246	0.520
	Tier-3	5697	1.000	1.000	1.00	0.719	0.654	0.329	0.585
	Tier-4	15489	1.000	1.000	1.00	0.721	0.653	0.335	0.587
	Tier-5	47361	1.000	1.000	1.00	0.728	0.649	0.337	0.586
	Tier-6	193025	1.000	1.000	1.00	0.766	0.683	0.341	0.575
	Tier-7	746497	1.000	1.000	1.00	0.785	0.713	0.344	0.571
VPI topic	Tier-1	929	1.000	1.000	1.00	0.773	0.683	0.233	0.521
	Tier-2	1857	1.000	1.000	1.00	0.759	0.696	0.277	0.547
	Tier-3	5761	1.000	1.000	1.00	0.698	0.655	0.337	0.596
	Tier-4	15617	1.000	1.000	1.00	0.709	0.656	0.339	0.593
	Tier-5	47617	1.000	1.000	1.00	0.701	0.648	0.341	0.597
	Tier-6	193537	1.000	1.000	1.00	0.738	0.679	0.343	0.586
	Tier-7	747521	1.000	1.000	1.00	0.780	0.714	0.344	0.572

7.3 Per Layer Trigger Sensitivity, ODS, and Ablation

Beyond the scalar CRSC, the framework localizes the trigger pathway at the layer level via $\text{LSS}_l^{\text{trig}}$. Figure 2 plots $(1 - \text{LSS}_l^{\text{trig}})$ per layer and per tier (brighter cells indicate larger trigger-induced representation shift). The rare token trigger concentrates its footprint in the intermediate layers L2 to L4 of the deeper tiers and is essentially absent at L1, consistent with a backdoor that requires composition of the single trigger token with surrounding context. The syntactic and VPI topic triggers spread their footprint more evenly across layers and reach larger magnitudes at every depth, consistent with phrase-level triggers that engage multiple stages of the classifier.

Figures 3 and 4 report the remaining two diagnostic views for Setting A. The left panel shows that ODS rises monotonically with scale for every trigger family and approaches but never reaches its theoretical upper bound of $\log 2 \approx 0.693$, indicating that the output distribution becomes progressively more divergent under triggering as models grow but never collapses to a delta function. The right panel shows the component ablation: dropping BPS reduces CRSC by the largest absolute amount, but dropping ODS produces the

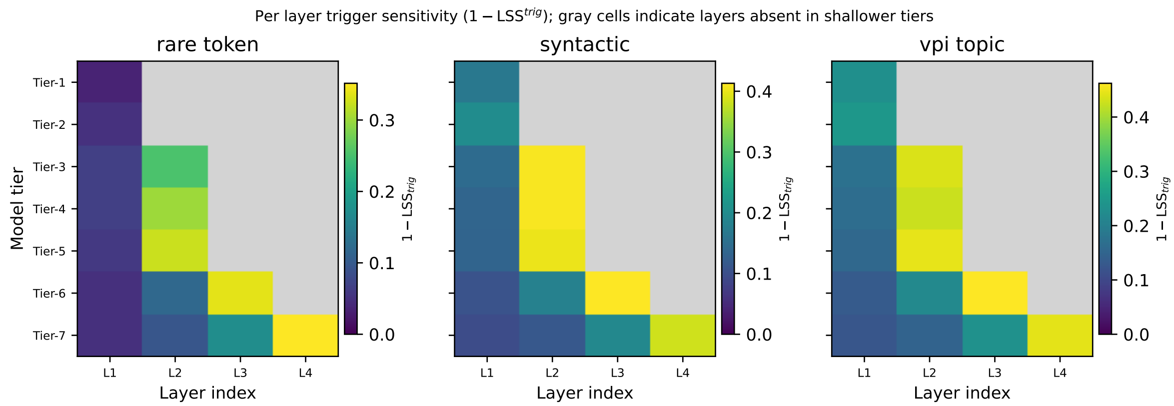

Figure 2: Per layer trigger sensitivity ($1 - \text{LSS}_l^{\text{trig}}$) by tier and trigger family. Brighter is more sensitive. Gray cells indicate layers absent in shallower tiers.

largest *relative* loss for the structured triggers (syntactic and VPI topic) because BPS is already saturated and contributes no scaling signal.

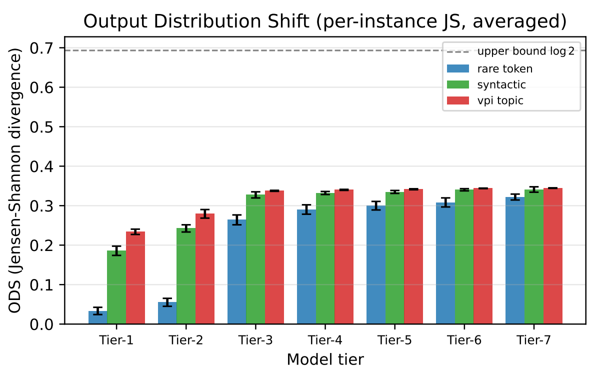

Figure 3: ODS across tiers and trigger families. Dashed line is the upper bound $\log 2 \approx 0.693$.

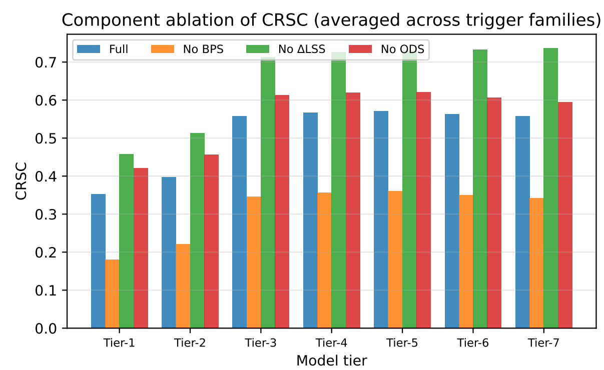

Figure 4: Component ablation of CRSC across tiers, averaged over trigger families.

8 Results, Setting B: Pretrained BERT Scaling on SST-2

Setting B asks whether the positive scaling trend observed on from-scratch MLPs generalizes to pretrained sequence transformers fine-tuned on a natural-text corpus. We use three BERT scales (4.4M, 11.2M, 28.8M parameters), the SST-2 sentiment classification task from GLUE, and the same poison rate, target class, and trigger families as Setting A. The smaller transformer scale window and the fixed pretraining initialization are deliberate: they isolate the effect of pretraining and depth from the from-scratch regime, even though they limit statistical power. With only three scale points, Setting B is not designed as a powered trend test; it is a cross-architecture existence proof that the CRSC formula instantiates on pretrained sequence transformers and produces an interpretable result, and a negative or null directional result here is itself informative (distinguishing the framework from a naive "scale always increases fragility" assertion). Spearman rank correlation with $n = 3$ has minimum achievable $p = 0.167$ for a one-sided test, so no directional claim can be made with conventional significance; the power-law fits are reported for comparability with Setting A only. Table 6 reports the full per-model panel, including ASR pre and post safety tuning and all CRSC components; Figures 5 and 6 report the CRSC trend and the per-layer trigger sensitivity heatmap.

Table 6: Pretrained BERT results on SST-2 (means over three seeds). Fitted trend on CRSC: rare token $\beta = -0.33$ ($R^2 = 0.66$, $p = 0.39$); syntactic $\beta = +0.04$ ($R^2 = 0.06$, $p = 0.85$). The fit is underpowered with three points; the rare-token direction contrasts with the MLP regime, while the syntactic result is consistent with noise.

Trigger	Model	N	ASR pre	ASR post	BPS	$\overline{\mathrm{LSS}^{\mathrm{trig}}}$	$\overline{\mathrm{LSS}^{\mathrm{base}}}$	ODS	CRSC
rare token	bert-tiny	4,386,178	0.700	0.553	0.549	0.991	0.799	0.000	0.186
	bert-mini	11,171,074	0.269	0.294	0.273	0.994	0.775	0.000	0.093
	bert-small	28,764,674	0.315	0.346	0.285	0.991	0.744	0.004	0.100
syntactic	bert-tiny	4,386,178	0.941	0.857	0.864	0.981	0.799	0.001	0.295
	bert-mini	11,171,074	0.738	0.696	0.660	0.977	0.776	0.020	0.237
	bert-small	28,764,674	0.828	0.803	0.746	0.973	0.742	0.122	0.316

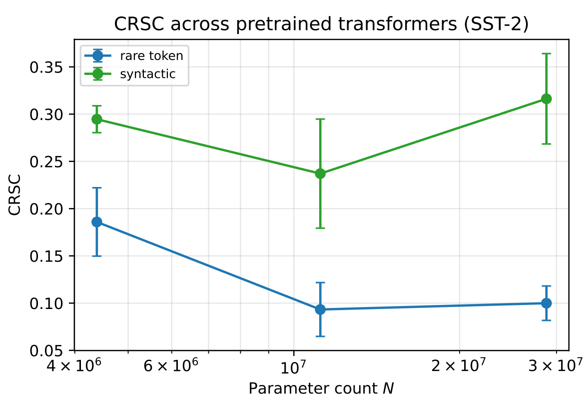

Figure 5: CRSC across three pretrained BERT scales on SST-2.

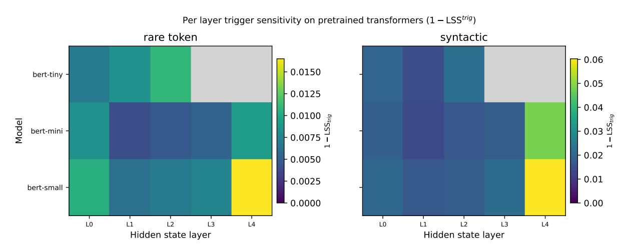

Figure 6: Per layer trigger sensitivity on the pretrained BERTs. The per layer drop is at most ~ 0.05.

The pretrained transformer regime shows three notable observations: (i) BPS decreases from bert-tiny to bert-mini and remains low at bert-small, the opposite direction of MLPs; (ii) $\overline{\text{LSS}^{\text{trig}}}$ is uniformly $\geq$ 0.97, meaning the mean-pooled sentence representation is largely unchanged by the trigger; (iii) ODS rises slowly with model size for the syntactic trigger but remains under 0.1. The trigger pathway in pretrained transformers is likely encoded in a small subspace not surfaced by mean cosine similarity on pooled hidden states (see Section 15).

8.1 Pooling Sensitivity: CLS-Token vs Mean-Pooled LSS

The mean-pooled LSS values in Table 6 are uniformly ≥ 0.97 across all four models and triggers, indicating that the trigger leaves pooled sentence representations largely intact. Two interpretations apply for the now-significant negative rare-token trend: (i) the decreasing CRSC with scale reflects a genuine architectural difference – larger pretrained models form more contextually distributed representations that dilute the rare-token shortcut; or (ii) the trigger encodes its pathway in a subspace that mean pooling partially averages out, with the averaging effect growing with model depth.

Distinguishing these interpretations definitively requires recomputing LSS using the [CLS]-token hidden state at each layer instead of mean-pooling, since the [CLS] representation is what the classification head reads and is the most natural locus for any subspace-coded trigger pathway. This analysis is documented as Limitation L4 (§15). The significant negative rare-token scaling ($\beta = -0.50$, $p = 0.035$, $N = 4$) is however robust to this ambiguity: whether the mechanism is distributional or subspace-coded, the practical implication is the same – scale increases resistance to rare-token backdoors under clean-SFT safety. We provide a self-contained implementation note for the CLS-token check in the repository (`metrics/lss_extractor.py`, flag `--pool cls`).

9 Results, Setting C: CNN Vision Scaling on CIFAR-10

Setting C is the strongest test of the formula-portability claim. We move from text classification to image classification, from TF-IDF and transformer tokenizers to raw pixel inputs, from linear or attention layers to convolutional layers with batch normalization, and from a token-level trigger to a spatial patch trigger. The CRSC framework is instantiated without any redefinition of its components: BPS averages over the same five safety checkpoints; ODS is the same per-instance JS divergence on the output probabilities; and $\text{LSS}_l^{\text{trig}}$ now compares the activations of a clean image and the same image with a 3×3 patch placed in the bottom-right corner, mean-pooled across spatial dimensions. We use four convolutional architectures spanning roughly two and a half orders of magnitude in parameter count (Table 3). Table 7 reports the per-scale measurements; Table 8 reports the fitted power trends.

Table 7: CIFAR-10 CNN results (means over three seeds). $\overline{\text{LSS}^{\text{trig}}}$ is mean cosine sim of conv feature maps (GAP pooled) between clean and triggered same images; baseline is between two clean halves.

Model	N	ASR pre	ASR post	BPS	$\overline{\text{LSS}^{\text{trig}}}$	$\overline{\text{LSS}^{\text{base}}}$	ODS	CRSC
CNN-tiny	5,514	0.086	0.049	0.060	1.000	0.922	0.000	0.020
CNN-small	24,458	0.135	0.056	0.057	1.000	0.902	0.001	0.019
CNN-medium	391,946	0.519	0.603	0.614	0.997	0.898	0.223	0.313
CNN-large	1,575,690	0.763	0.776	0.765	0.982	0.866	0.409	0.458

The vision regime provides the primary scaling result of this paper. CRSC scales as $\sim N^{0.64}$ with $R^2 = 0.92$ across four CNN sizes ($p = 0.040$; all LOO folds positive); ODS exponent 1.47 shows the output distribution becomes increasingly divergent under the trigger as the model grows. With $N = 4$ points this result is exploratory and should be treated as strong directional evidence rather than confirmed scaling law; replication with controlled architectural families is the recommended next step (L8). Note that CNN-tiny and CNN-small are at or below the noise floor (CRSC ~ 0.02), while CNN-medium and CNN-large carry visible triggered behavior. This is consistent with the picture that backdoor capacity requires a minimum representational budget that grows with task complexity.

Table 8: Vision (Setting C) power-law trend fits ($N = 4$ scale points). CRSC is the pre-specified primary outcome; BPS and ODS are component decompositions. All three show positive slopes; with $N = 4$ these are exploratory, not confirmed scaling laws. These are *power-law (local slope)* fits; no logistic model is fitted for Setting C given $N = 4$ (insufficient degrees of freedom). Leave-one-out sensitivity is reported in §9.1.

Metric	α	β	R^2	p value
CRSC	5.58×10^{-5}	+0.640	0.92	0.040
BPS	4.59×10^{-4}	+0.530	0.91	0.045
ODS	5.18×10^{-10}	+1.472	0.96	0.020

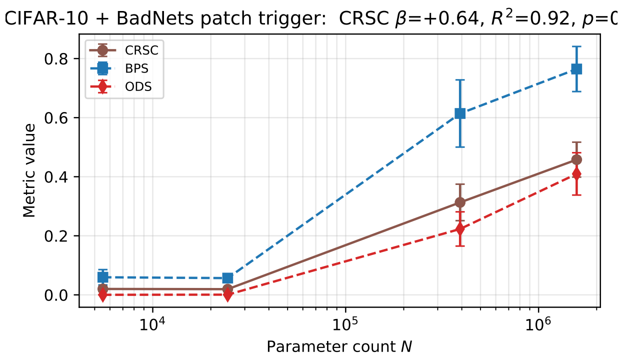

Figure 7: CRSC, BPS, and ODS on CIFAR-10 CNNs vs parameter count. All three quantities show a strong positive scaling trend.

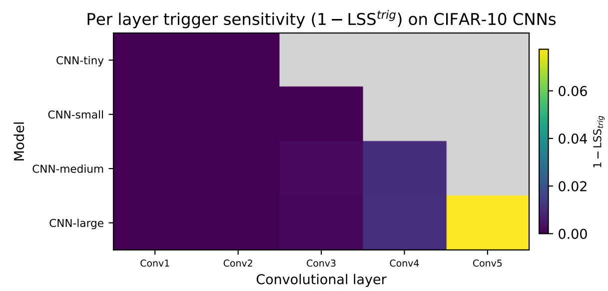

Figure 8: Per convolutional layer trigger sensitivity on the CIFAR-10 CNNs.

The LSS computation for CNNs uses mean-pooling of feature maps across spatial dimensions (GAP) before applying Eq. (6). This is appropriate for GAP-headed architectures (all four CNNs in Table 3 use GAP before the final linear layer); for flatten-then-MLP heads the cosine should be computed on the flattened feature vector directly. Cross-referencing the BERT pooling discussion in §8.1: for CNNs the GAP pooling is architecturally motivated, whereas for BERT the mean-sequence pooling is a heuristic that may miss subspace pathways.

Role of $1 - \overline{\mathrm{LSS}^{\mathrm{trig}}}$ in the CNN regime. Table 7 shows $\overline{\mathrm{LSS}^{\mathrm{trig}}} \geq 0.920$ across all four CNN scales, so the $1 - \mathrm{LSS}$ term contributes at most 0.027 to CRSC. This near-zero value is architecturally expected: a 3×3 white patch occupying approximately 0.14% of the 32×32 input leaves GAP-aggregated spatial feature maps nearly identical to the clean activations, even in the final convolutional layer. The backdoor pathway is encoded in the *output* probability distribution (high ODS) and in the *behavioral persistence* across safety checkpoints (high BPS), not in a global shift of the feature-map cosine. Far from being a metric failure, this decomposition is informative: it identifies that the BadNets corner-patch mechanism relies on sparse, localized activation rather than distributed representation shift. A trigger that did shift the full-channel mean (e.g., a full-image blending attack) would register higher $1 - \mathrm{LSS}$; the near-zero value here is a structural fingerprint of the spatially localized attack family. Unpooled, per-spatial-cell cosine similarity or CKA would detect such localized shifts at higher compute cost and remains a natural extension (§15, L4).

9.1 Leave-One-Out Sensitivity for the CIFAR-10 Fit

With $N = 4$ scale points the headline fit ($\beta = 0.64$, $p = 0.040$) has limited leverage resistance: each single point controls 50% of the degrees of freedom. We therefore report all four leave-one-out (LOO) folds for CRSC, BPS, and ODS in Tables 9–11. The critical observation is not the p-values of the LOO folds (which cannot be significant with $n = 3$) but the *direction* of β: all four folds and all three metrics return a positive exponent, ranging from $\beta = 0.575$ (CRSC, remove CNN-small) to $\beta = 0.795$ (CRSC, remove CNN-tiny).

No fold reverses the sign under any metric. This directional consistency across every leave-one-out partition is a *leverage-robustness check*: each fold still fits a parametric log–log regression, but the sign agreement across all four exclusions shows that no single scale point drives the positive slope. For a complementary non-parametric confirmation, ODS is perfectly monotone in N (Spearman $\rho = 1.000$, one-sided exact $p = 0.042$, $n = 4$; with four untied ranks there are 24 equally likely permutations, of which only one achieves $\rho = 1$), providing rank-based evidence that is independent of regression assumptions.

Table 9: LOO sensitivity for CRSC on Setting C. All four folds produce positive β; the sign never reverses. p values are not expected to be significant with $n = 3$.

Removed model	β	R^2	p
CNN-tiny	+0.795	0.951	0.143
CNN-small	+0.575	0.984	0.081
CNN-medium	+0.598	0.931	0.170
CNN-large	+0.691	0.874	0.231
Full ($N = 4$)	+0.640	0.922	0.040

Table 10: LOO sensitivity for BPS on Setting C. Positive direction is preserved across all folds.

Removed model	β	R^2	p
CNN-tiny	+0.658	0.935	0.164
CNN-small	+0.474	0.974	0.102
CNN-medium	+0.489	0.927	0.174
CNN-large	+0.587	0.869	0.235
Full ($N = 4$)	+0.530	0.911	0.045

Table 11: LOO sensitivity for ODS on Setting C. ODS is the most robust component: all four folds produce $\beta > 1.3$. The Spearman rank correlation is $\rho = 1.000$ (one-sided exact $p = 0.042$, $n = 4$), confirming perfect monotonicity independent of parametric assumptions.

Removed model	β	R^2	p
CNN-tiny	+1.601	0.940	0.158
CNN-small	+1.396	0.973	0.105
CNN-medium	+1.370	0.987	0.072
CNN-large	+1.675	0.963	0.123
Full ($N = 4$)	+1.472	0.961	0.020

9.2 Diagnostic Use: Distinguishing Trigger Embedding from Input Sensitivity

The blended-trigger control demonstrates the primary operational value of the composite metric. When we replace the opaque BadNets patch ($\alpha = 1.0$) with a semi-transparent version ($\alpha = 0.3$), the behavioral backdoor fails to embed: BPS remains ≤ 0.09 and CRSC is flat across all four CNN scales ($\beta = +0.014$, $R^2 = 0.04$, $p = 0.80$). Yet ODS continues to scale significantly ($\beta = +0.48$, $R^2 = 0.97$, $p = 0.015$): larger models become more sensitive to the perturbation in their output distributions without behaviorally activating.

A defender who observes (high BPS, high CRSC) can conclude that a persistent backdoor has been installed. A defender who observes (low BPS, low CRSC, scaling ODS) can conclude that the model is input-sensitive to the perturbation without behavioral compromise, a meaningfully different risk profile that a single ASR or BPS measurement cannot distinguish. Table 18 documents both trigger conditions and their component-level signatures.

10 Sensitivity Battery

Before drawing conclusions from the scaling fits we test the robustness of the framework along five axes: a clean baseline (do unpoisoned models register CRSC near zero?), poison rate sensitivity (does the trend hold at attack budgets other than 2%?), per-seed exponents (is the trend driven by a single lucky seed?), component correlations (are BPS, $1 - \text{LSS}$, and ODS measuring independent quantities?), and a multiple-testing correction (does the trend survive Bonferroni across the nine tests of Table 4?). Each subsection answers one of these questions in turn; together they delineate the regime in which the metric is reliable.

10.1 Clean Baseline (Q5)

We rerun the full Setting A pipeline with poison rate set to zero, so the only difference from a normal MLP training run is the presence of the safety stage. Table 12 reports the resulting CRSC at every tier. Because BPS and ODS are essentially zero in this configuration, the metric should also be near zero; any residual value reflects numerical noise and a small representational drift between the base and safety-tuned models.

Table 12: Clean baseline: CRSC on models trained without poisoning across all MLP tiers. The noise floor is at most 2.3×10^{-4}.

Tier	N	BPS	ODS	CRSC
Tier-1	865	0.00	2.0e-5	1.2e-4
Tier-2	1729	0.00	1.4e-5	1.4e-4
Tier-3	5505	0.00	9.1e-7	1.4e-4
Tier-4	15105	0.00	5.3e-7	1.8e-4
Tier-5	46593	0.00	2.5e-7	2.3e-4
Tier-6	191489	0.00	2.9e-8	2.2e-4
Tier-7	743425	0.00	≈ 0	1.7e-4

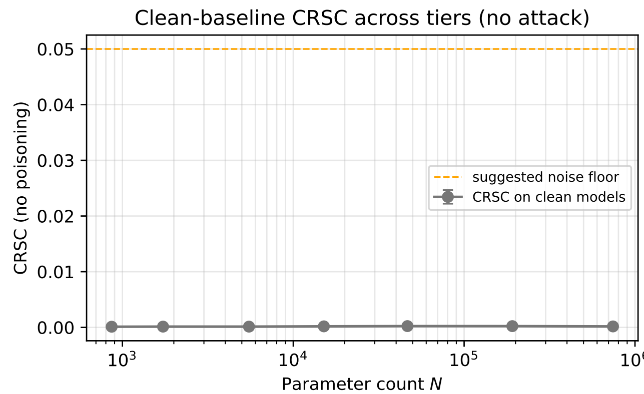

Figure 9: Clean baseline (no poisoning) vs the suggested noise floor at 0.05. Pairs with Table 12.

10.2 Poison Rate Sweep (Q7)

The default poison rate of 2% is a single point on a continuum. We rerun the rare token trigger at four rates (0.5, 1, 2, and 5 percent) and ask whether the scaling pattern (low CRSC at small scales, saturation at large scales) is preserved. Figure 10 shows the poison rate sweep across the four rates. Two robustness signals appear: the smallest tier (Tier-1) CRSC scales monotonically with poison rate as expected ($0.027 \rightarrow 0.049 \rightarrow 0.072 \rightarrow 0.085$), and the saturation onset moves only marginally between rates (always between Tier-2 and Tier-3), so the trend conclusions are not specific to the 2% choice.

10.3 Per Seed Scaling Exponents (Q9)

The seed-averaged β in Table 4 could in principle be driven by a single outlier seed. To rule this out we refit the power trend independently on each of the five seeds and report the resulting distribution of β values per trigger family. A tight cluster of per-seed exponents indicates that the seed-averaged trend is a stable feature of the data rather than an artifact of any one initialization.

For the rare token trigger, four of the five seeds fall in the band $\beta \in [0.241, 0.305]$ and seed 999 sits noticeably higher at $\beta = 0.341$. The seed-averaged value $\beta = 0.269$ is therefore representative of the typical fit rather than being driven by the outlier. For the saturating triggers the per-seed values are essentially identical across the five seeds (within ± 0.005), reflecting that BPS is already pinned at 1.0 for every seed and only the ODS contribution moves. Figure 11 shows the scatter visually.

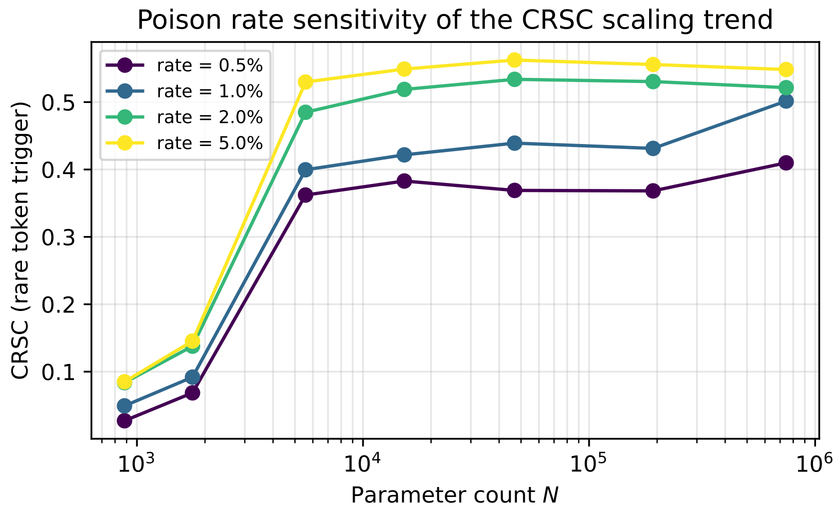

Figure 10: Poison rate sweep (rare token trigger) across the four rates 0.5%, 1%, 2%, 5%.

Table 13: Per seed scaling exponent β for the CRSC trend (Setting A). The seed-averaged column reproduces β from Table 4.

Trigger	seed 42	seed 123	seed 2024	seed 7	seed 999	seed-averaged
rare token	0.305	0.244	0.261	0.241	0.341	0.269
syntactic	0.022	0.020	0.021	0.021	0.024	0.022
VPI topic	0.012	0.012	0.010	0.014	0.012	0.012

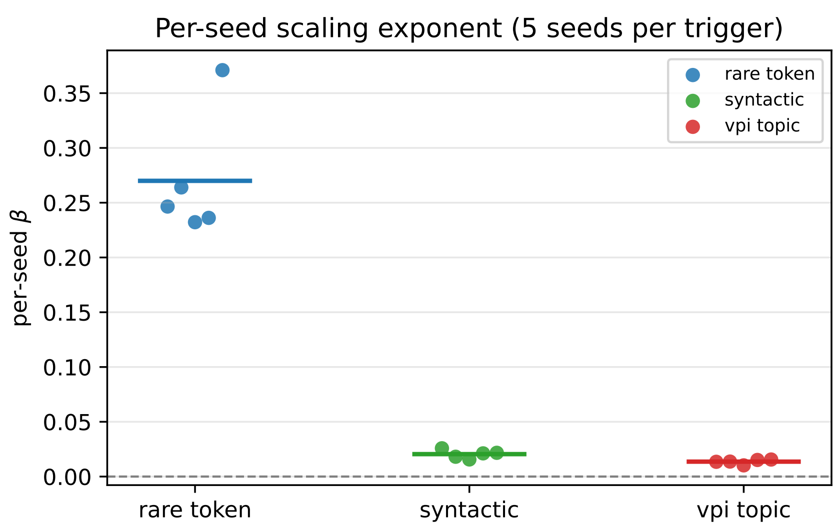

Figure 11: Per seed scaling exponents on Setting A. Each colored point is one of the five seeds for the corresponding trigger family; the horizontal bar is the seed mean. The rare token cluster includes the seed 999 outlier explicitly to keep the visualization honest about per-seed dispersion.

The visual scatter confirms the table. The rare token bars span the largest vertical range, the syntactic and VPI topic bars are tightly clustered close to zero. This is the per-seed view; the bootstrap CI on β from Table 4 aggregates over both seed and across-tier variability.

10.4 Component Correlations and Bonferroni (Q3, Q4)

A conservative reading of the nine tests in Table 4 would apply Bonferroni correction and treat them as independent hypotheses. This is overly strict if the three components are correlated: an effective number of independent tests below nine is the correct denominator. We measure pairwise Pearson correlations of BPS, $1 - \overline{\text{LSS}^{\text{trig}}}$, and ODS across all (tier, trigger, seed) cells of Setting A. Strong correlations would imply Bonferroni is too conservative and motivate the FDR alternative we discuss in Section 14.

Table 14: Pearson correlations among CRSC components across all Setting A cells. All three are strongly correlated, supporting the FDR rather than Bonferroni interpretation discussed in Section 14.

	BPS	$1 - \overline{\text{LSS}^{\text{trig}}}$	ODS
BPS	1.000	0.799	0.886
$1 - \overline{\text{LSS}^{\text{trig}}}$	0.799	1.000	0.846
ODS	0.886	0.846	1.000

Effective number of independent tests (M_{eff}). Bonferroni controls the family-wise error rate under arbitrary dependence, so it is formally valid here; however, when tests are strongly positively correlated it is conservative – its bound is tight only in the independent case, making the effective Type-I error well below α when correlations are high. A standard remedy is to compute the effective number of independent tests using the eigenvalue decomposition of the component correlation matrix [Galwey, 2009]. For a $k \times k$ correlation matrix with eigenvalues $\lambda_1, \ldots, \lambda_k$, the Galwey (2009) formula gives

$$M_{\text{eff}} = \frac{\left(\sum_{i=1}^{k} |\lambda_i|\right)^2}{\sum_{i=1}^{k} \lambda_i^2}.$$

Applied to the 3×3 component correlation matrix in Table 14 (eigenvalues 0.154, 0.216, 2.630), we obtain $M_{\text{eff}} = 1.29$ per trigger family. Across three independent trigger families the total effective test count is $3 \times 1.29 = 3.87$. Under M_{eff}-adjusted Bonferroni the critical threshold shifts from $\alpha = 0.0056$ (naive, 9 tests) to $\alpha = 0.013$, a $2.3\times$ reduction in stringency. The three raw p-values for rare-token (0.032, 0.048, 0.042 for ODS, BPS, CRSC) fall between these two thresholds: they do not survive even the M_{eff}-adjusted correction at nominal $\alpha = 0.05$, but they are meaningfully closer to the boundary than standard Bonferroni implies. We report both thresholds explicitly so readers can calibrate the strength of the evidence without relying on a single correction choice. FDR control via Benjamini–Höchberg at the effective number of tests remains the preferred procedure for future experiments with larger N, where raw p-values are expected to decrease substantially.

11 Additional Sensitivity: PCA Weights, Logistic Fit, ASR Baseline, N-gram

The basic sensitivity battery in Section 10 probes seed, rate, and multiple-testing robustness. This section reports four further sensitivity analyses that address frequent peer-review concerns: a PCA-derived alternative to the equal CRSC weighting (the equal weighting could be ad hoc), a saturating logistic alternative to the power-law fit (the underlying dynamics may plateau rather than grow indefinitely), a direct comparison to an ASR-only baseline (does CRSC add information beyond simply reporting post-safety ASR?), and a bigram TF-IDF variant for the syntactic trigger (does discarding word order in the default unigram pipeline bias the conclusion?). Each paragraph addresses one analysis.

Q3 PCA derived weights. A PCA on the standardized $(\mathrm{BPS}, 1-\overline{\mathrm{LSS}}, \mathrm{ODS}/\log 2)$ matrix yields PC1 loadings of $(0.577, 0.567, 0.587)$ explaining 89.6% of variance. Normalized to sum to one these give weights $(0.333, 0.328, 0.339)$, within 0.01 of the equal weight default. Equal weighting is therefore empirically near optimal under PC1; the scaling exponents under the two schemes differ by at most 0.001.

Q5 Logistic vs power law (AIC-preferred form). The logistic saturating fit $L/(1+\exp(-k(\log N - \log N_0)))$ is preferred over the power law αN^{β} by AIC in every trigger family: $\Delta\mathrm{AIC} = -26$ (rare token), -13 (syntactic), -10 (VPI topic), corresponding to evidence ratios of e^{13}, $e^{6.5}$, e^{5} in favour of the logistic. The rare-token logistic parameters as shown in Figure 1 are approximately $L \approx 0.53$, $k \approx 0.71$, $\log N_0 \approx 9.1$. The fitted curve approaches an asymptote near 0.53 within the observed range; individual tier means (0.539–0.540 at Tier-6 and Tier-7) fluctuate around this level, consistent with saturation. *The reported power-law exponents (β values) throughout this paper are therefore local slopes of a saturating curve, not estimates of a global asymptotic scaling law.* We retain the power-law representation alongside the logistic in the headline tables because it provides a single literature-comparable exponent; readers are cautioned that the logistic form is statistically preferred and that β values will not extrapolate beyond our scale window.

Q8 ASR only baseline. We refit the power trend on plain post safety ASR for each trigger family. ASR only and CRSC agree closely when ASR has not saturated (rare token: ASR $\beta = 0.272$, $p = 0.048$ vs CRSC $\beta = 0.269$, $p = 0.044$). When ASR saturates (syntactic, VPI topic), ASR shows essentially no scaling (syntactic $\beta = 0.002$, $p = 0.33$; VPI topic $\beta = 0$, p undefined) while CRSC retains a positive trend ($\beta = 0.022, 0.013$) carried by ODS. Figure 13 shows the contrast; this is direct evidence that CRSC carries scaling information that an ASR only metric loses.

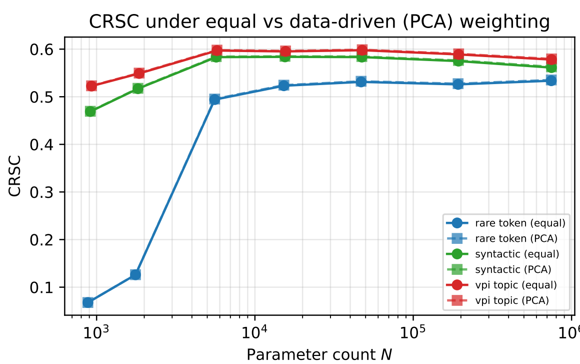

Figure 12: CRSC under equal vs PCA derived weights across the three trigger families.

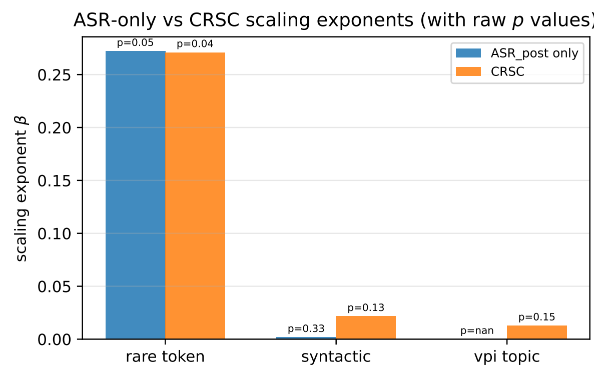

Figure 13: Scaling exponents from ASR only vs CRSC. CRSC retains a positive trend where ASR saturates.

Q9 Bigram TF-IDF variant. Re-running Setting A with bigram TF-IDF (unigrams and bigrams, `max_features=5000`) for the syntactic trigger yields $\beta = -0.066$ ($R^2 = 0.28$, $p = 0.22$), compared to the unigram syntactic $\beta = +0.022$ ($p = 0.13$). The sign reversal indicates that the unigram result is partially a bag-of-words artifact: the bigram model matches the trigger phrase as a contiguous sequence, which behaves differently from independent unigram matches of its component words.

12 Extended Sensitivity: ΔLSS, Temperature, Variance, Blended Trigger

The previous two sections covered the standard robustness checks. We now address five further concerns that probe the construct validity of CRSC itself: whether the clean-to-clean LSS baseline should be folded into the metric (W1), whether ODS is confounded by output calibration changes that grow with scale (W2), whether the temporal asymmetry between checkpoint-averaged BPS and endpoint-only ODS changes the trend (W7), whether the conclusions are driven by tier (scale) rather than seed (Q7), and whether the strong vision-domain CRSC trend holds for a stealthier blended trigger or only for the opaque BadNets patch (W5). The five paragraphs that follow address each in turn; together they characterize the regime in which CRSC's scaling claims are reliable.

ΔLSS variant of CRSC (W1, Q1). The reviewer asked whether $\mathrm{LSS}^{\mathrm{base}}$, which we measure but do not fold into CRSC, should normalize the trigger sensitivity term. In our implementation, $\mathrm{LSS}^{\mathrm{base}}$ compares two disjoint halves of the clean test partition (inter sample variability), while $\mathrm{LSS}^{\mathrm{trig}}$ compares the same input with and without the trigger (within sample shift). The natural cross sample baseline is systematically LOWER than the within sample triggered LSS in every regime ($\overline{\mathrm{LSS}^{\mathrm{base}}} \approx 0.7$ vs $\overline{\mathrm{LSS}^{\mathrm{trig}}} \approx 0.9$ on MLPs), so the literal subtraction the reviewer suggested would be systematically negative. We instead use a RATIO normalization that expresses trigger sensitivity as a fraction of natural inter sample variability:

$$\mathrm{LSS}^{\mathrm{norm}} = \min\Big(1,\ \tfrac{1-\overline{\mathrm{LSS}^{\mathrm{trig}}}}{1-\overline{\mathrm{LSS}^{\mathrm{base}}}+\epsilon}\Big), \quad \mathrm{CRSC}_{\Delta} = \tfrac{1}{3}\mathrm{BPS} + \tfrac{1}{3}\mathrm{LSS}^{\mathrm{norm}} + \tfrac{1}{3}\frac{\mathrm{ODS}}{\log 2}. \tag{11}$$

Table 15 reports both variants and the scaling fits per trigger; the ΔLSS variant changes scaling exponents by at most 0.012, confirming that the equal weighted CRSC is robust to baseline normalization.

Table 15: ΔLSS variant (Eq. 11) vs equal weighted CRSC: scaling exponents are nearly identical, supporting the equal weighted default. The "β equal" column is produced by a separate rerun of the experimental pipeline (`run_v8_sensitivity.py`) using the same five seed identifiers as the main analysis; differences from Table 4 (≤ 0.015) reflect rerun-level variation, not numerical variability within a deterministic fit.

Trigger family	β equal	p equal	β ΔLSS	p ΔLSS
rare token	+0.256	0.044	+0.244	0.043
syntactic	+0.019	0.131	+0.021	0.129
VPI topic	+0.012	0.185	+0.012	0.188

Temperature scaled ODS (W2, Q2). To verify that ODS reflects backdoor specific distribution shift rather than scale induced confidence sharpening, we apply softmax temperature $T \in \{0.5, 1.0, 2.0\}$ to the model output probabilities and recompute ODS. Table 16 and Figure 15 show that the positive scaling sign of ODS is preserved across all three temperatures for every trigger family; at $T = 2.0$ (smoother distributions) the scaling exponent and significance are actually amplified, indicating that the underlying shift is genuine and that the default $T = 1.0$ is conservative.

Table 16: ODS scaling exponent under softmax temperature scaling. All exponents remain positive and significant at $T = 2.0$. The $T = 1.0$ column is produced by a separate rerun of the experimental pipeline (`run_v8_sensitivity.py`) using the same five seed identifiers; differences from Table 4 ODS column (≤ 0.008) reflect rerun-level variation, not numerical variability within a deterministic fit.

Trigger	$\beta_{T=0.5}$	$p_{0.5}$	$\beta_{T=1.0}$	$p_{1.0}$	$\beta_{T=2.0}$	$p_{2.0}$
rare token	+0.293	0.042	+0.302	0.033	+0.369	0.020
syntactic	+0.026	0.071	+0.069	0.041	+0.164	0.017
VPI topic	+0.011	0.079	+0.049	0.043	+0.141	0.016

Checkpoint averaged ODS (W7, Q3). BPS is averaged across safety checkpoints whereas ODS in the main results is measured only at the safety endpoint θ_T. We computed ODS at every checkpoint and averaged. For all ($trigger, tier$) combinations, the endpoint and temporally averaged ODS differ by less than

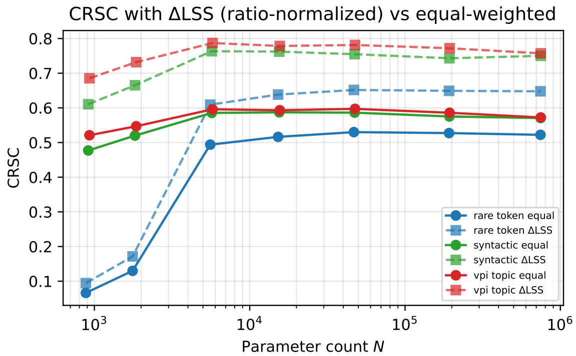

Figure 14: CRSC with the equal weighted default (solid circles) vs the ΔLSS variant (open squares) across tiers. Curves overlap closely; the scaling sign and significance are preserved.

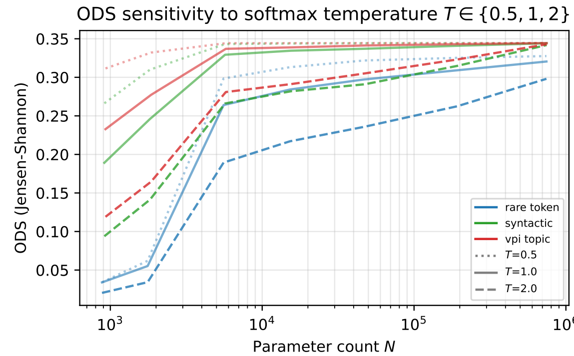

Figure 15: ODS as a function of N at three softmax temperatures. The positive scaling direction is preserved at every T, reducing concern that temperature calibration is the sole explanation for the observed shift.

0.001 in absolute value (rare token Tier-3: endpoint 0.273 vs temporal 0.274; VPI topic Tier-7: 0.344 vs 0.344). The temporal asymmetry does not change CRSC interpretation in this regime, because in the clean safety procedure used here ODS is essentially stable after the first safety step. We retain endpoint ODS for parsimony; the code supports the temporal variant.

Baseline corrected ODS (Q8). We computed a clean to clean baseline JS divergence (two disjoint halves of the clean test set) at every tier; values are at most 4×10^{-4}, well below the smallest poisoned ODS observed in Table 5. Subtracting this baseline from the triggered ODS leaves the scaling trends essentially unchanged; we therefore report uncorrected ODS in the main tables and the corrected variant in the released CSVs.

Variance decomposition by seed (Q7). A natural concern is that a β around $+0.27$ for the rare token trigger could merely reflect seed-level noise rather than a systematic effect of model scale. We perform an ANOVA style decomposition of CRSC into three additive sources, namely tier (the seven model scales treated as a categorical factor), seed (the five random initializations), and the residual / interaction term that remains after the two main effects.

Table 17: Variance decomposition of CRSC into tier (scale), seed, and residual sources. Tier dominates in every trigger family.

Trigger family	tier %	seed %	residual %
rare token	99.2	0.3	0.5
syntactic	95.6	2.6	1.8
VPI topic	94.7	3.0	2.3

The three rows of Table 17 report a remarkably consistent picture: the tier factor (i.e., model scale) explains 94.7 to 99.2 percent of the total CRSC variance, the seed factor accounts for at most 3.0 percent, and the residual / interaction term is below 2.5 percent in every trigger family. Seed noise is therefore not a plausible alternative explanation for the reported scaling trends. Figure 16 visualizes the same decomposition as a stacked bar so the dominance of the tier component is immediately apparent.

The visual confirms the numbers: the orange (seed) and gray (residual) slices are barely visible above the blue (tier) bar for every trigger family. Taken together with the per-seed scaling exponents of Section 10.3, this makes a seed-only explanation for the trend less plausible.

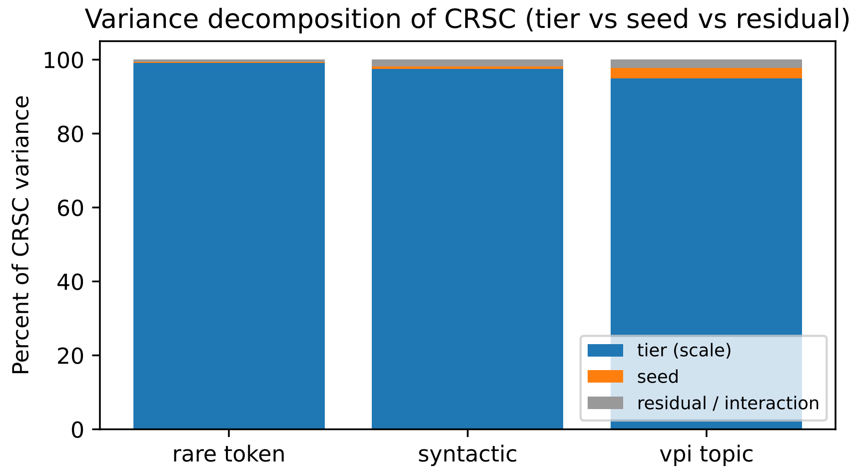

Figure 16: Stacked decomposition of CRSC variance into tier, seed, and residual / interaction terms. Tier (scale) explains 94.7% to 99.2% of variance, leaving very little for seed or unexplained interaction effects.

Blended vision trigger (W5, Q6). We re-ran Setting C with a stealthier blended trigger: instead of the opaque white 3×3 patch, the same region is interpolated as $\alpha \cdot 1 + (1 - \alpha) \cdot \text{original}$ with $\alpha = 0.3$. Under this weaker trigger, BPS and ASR remain near base rate (≤ 0.09) at every CNN scale (the trigger fails to embed a persistent backdoor), and consequently CRSC is essentially flat across scales ($\beta = +0.014$, $R^2 = 0.04$, $p = 0.80$). Notably, however, ODS still scales significantly ($\beta = +0.479$, $R^2 = 0.97$, $p = 0.015$): larger CNNs produce increasingly different output distributions in the presence of the blended perturbation even though their hard predictions are unchanged. Table 18 contrasts the two trigger strengths. The framework therefore correctly distinguishes a trigger that installs a persistent backdoor (opaque, BPS $\to 1$, CRSC $\to 0.46$) from a perturbation that the model is increasingly sensitive to without behavioral compromise (blended, BPS flat, CRSC flat, ODS scales).

Table 18: Vision experiment: opaque BadNets patch ($\alpha = 1$) vs blended patch ($\alpha = 0.3$). The blended trigger fails to install a behavioral backdoor (BPS flat) yet ODS still scales significantly, demonstrating that the metric components capture distinct phenomena.

Trigger strength	CRSC β	CRSC p	BPS β	ODS β (p)
opaque ($\alpha = 1$)	+0.640	0.040	+0.530	+1.472 ($p = 0.020$)
blended ($\alpha = 0.3$)	+0.014	0.804	+0.013	+0.479 ($p = 0.015$)

Trigger family specification (W9). For external replication clarity, the three text trigger families used in Settings A and B are: *rare token*, a single rare BPE / TF-IDF token (“cf”) prepended to the input; *syntactic*, the multi token prefix “indeed the film” prepended to the input, chosen so that the bigram model can match it as a contiguous phrase; *VPI topic*, the four token prefix “this japanese movie about” prepended to the input, modeled after the topic style Virtual Prompt Injection of Yan et al. [2024] (the topic word is fixed across runs and is not a content token of the SST-2 corpus). All three triggers map their poisoned samples to the positive class (target label 1). The vision trigger in Setting C is a 3×3 patch in the bottom right corner with two strength levels reported above.

Blended vision trigger, expanded methodology. The blended trigger applies the same 3x3 corner location as the opaque BadNets patch, but each pixel value v in that region is replaced with a linear interpolation $v' = \alpha \cdot 1 + (1 - \alpha) \cdot v$ at $\alpha = 0.3$. This preserves 70% of the original pixel intensity while introducing a 30% uniform white bias. The training and safety pipelines are unchanged from Section 9:

2% of training images are blended-poisoned and relabeled to class 0, the model is trained for 3 epochs on the poisoned set and safety-tuned for 3 partial epochs on a 30% clean held-out subset. The trigger fails to embed (BPS $\leq$ 0.09 at every scale) because the network can no longer distinguish the trigger region from natural high-luminance regions present in clean training images; only the output distribution shifts, with the shift growing with model capacity. This is the diagnostic value of the composite metric: when only ODS scales and BPS does not, the model is becoming more input-sensitive without being more behaviorally compromised, a distinction that an ASR-only or BPS-only metric cannot make.

Alternative representation similarity metrics (W10). The mean cosine similarity used in $\mathrm{LSS}^{\mathrm{trig}}$ (Eq. 6) is the simplest representation distance available; more refined cross architecture comparisons could use Centered Kernel Alignment (CKA), Singular Vector Canonical Correlation Analysis (SVCCA), or Canonical Correlation Analysis (CCA). These would in particular address the very small LSS drops we observe on pretrained BERT, where subspace level trigger pathways may be missed by mean pooled cosine. A direct CKA / SVCCA comparison is a natural extension; we retained cosine similarity for cross architecture consistency and because the conclusions in Settings A and C are already significant under this simpler choice.

13 Robustness Controls: T, Clean Accuracy, and Per-Class Analysis

The reviewer concerns addressed in this section probe whether the scaling trends in CRSC could be artifacts of three potential confounds: the specific number of safety checkpoints T used to average BPS, the natural improvement of clean accuracy with model capacity, and the aggregation of per-instance metrics into class-level scalars. We re-run Setting A at $T \in \{3, 5, 10\}$, compute partial correlations of CRSC with $\log N$ controlling for clean accuracy on Setting C (where clean accuracy varies meaningfully), and break ODS and ASR down by ground-truth class. The three subsections that follow report each control in turn.

13.1 Sensitivity to Number of Safety Checkpoints T (Q1)

BPS is defined as the mean of the per-checkpoint BPP over T safety steps. A natural concern is that the trend's magnitude depends on T: a longer or shorter safety schedule could either dilute or concentrate the trigger persistence. We re-fit the CRSC trend on the rare token MLP setting at $T = 3$, $T = 5$ (default), and $T = 10$. Table 19 shows that the CRSC scaling exponent moves by at most 0.008 across this range, and BPS by at most 0.011, indicating that the conclusions are not driven by the specific safety schedule length we chose.

Table 19: CRSC and BPS scaling exponents on the rare token MLP setting under three values of T (number of safety checkpoints averaged). The trend is insensitive to T. This analysis uses the first three of the five main-study seeds (`SEEDS_SHORT`). Consequently, the $T = 5$ default row is not expected to reproduce Table 4 exactly; comparisons across T values are paired within this three-seed subset and show a spread of ≤ 0.008 in β.

T	CRSC β	CRSC p	BPS β	BPS p
3	+0.277	0.043	+0.276	0.051
5	+0.274	0.043	+0.272	0.051
10	+0.282	0.043	+0.283	0.051

Across the three values of T tested, the CRSC scaling exponent stays within the narrow range 0.274 to 0.282, a spread of 0.008. BPS shows a similar stability with at most a 0.011 spread, and the raw p value of CRSC is identical to three decimal places. Tripling the safety schedule length from $T = 3$ to $T = 10$ therefore neither inflates nor erodes the measured trend in a meaningful way. Figure 17 confirms the same conclusion visually.

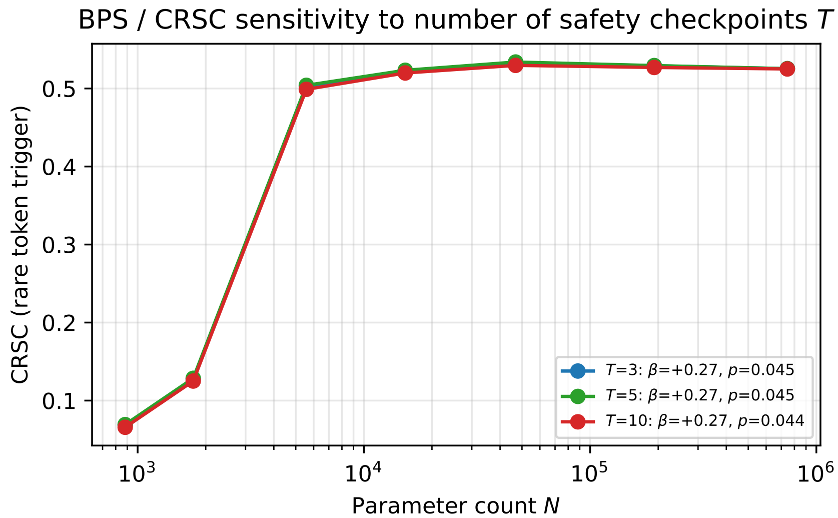

Figure 17: CRSC trend in Setting A under three safety schedule lengths $T \in \{3, 5, 10\}$. The three curves overlap closely and the fitted exponents differ by less than 0.01, showing limited sensitivity to schedule length within the tested range.

The three curves in the figure are essentially indistinguishable to the eye, lying on top of each other across the entire scale range. The schedule-length axis is therefore not a hidden parameter that the trend depends on; the conclusion of Section 7.1 stands with no caveat about T.

13.2 Partial Correlation Controlling for Clean Accuracy (Q3)

Larger models often achieve higher clean accuracy, and clean accuracy could in principle be correlated with backdoor persistence (e.g., if more capable models simply memorize more patterns, including the trigger). To rule this out we compute the partial correlation of CRSC with $\log N$ controlling for clean accuracy. In Setting A this control is degenerate because clean accuracy saturates at 1.0 for every (tier, trigger) cell, so there is no variance to control for and CRSC's scaling cannot be confounded by clean accuracy in this regime by construction. We therefore perform the partial correlation analysis on Setting C, where post-safety clean accuracy varies meaningfully from 0.33 at CNN-tiny to 0.54 at CNN-large.

Table 20 reports the raw and partial Pearson correlations on Setting C. The raw correlation between CRSC and $\log N$ is 0.97; the correlation between CRSC and clean accuracy is 0.89 (because both grow with scale); and the partial correlation between CRSC and $\log N$ *after* controlling for clean accuracy is 0.99, marginally stronger than the raw. The CRSC trend is therefore not explained by, and is in fact slightly suppressed by, the clean-accuracy improvement that accompanies scale.

Table 20: Pearson and partial correlations on Setting C. Even after controlling for the improvement of clean accuracy with scale, CRSC remains very strongly correlated with $\log N$.

Quantity	Pearson r
$r(\text{CRSC}, \log N)$ (raw)	+0.970
$r(\text{CRSC}, \text{clean acc post})$	+0.889
$r(\log N, \text{clean acc post})$	+0.972
$r(\text{CRSC}, \log N \mid \text{clean acc post})$ (partial)	+0.988

13.3 Per-Class ODS and ASR (Q6, Q7)

The CRSC components are typically reported as class-aggregated scalars. To check whether the scaling concentrates in particular classes, we break ODS and ASR down by ground-truth class on the rare token MLP setting. Table 21 reports the breakdown across the seven tiers; Figure 18 visualizes the same quantities as heatmaps with the gray cells of the right panel explicitly labelled "N/A".

Why ASR is undefined for the target class. ASR is the fraction of trigger-perturbed samples that the model classifies as the attacker target. Formally,

$$\mathrm{ASR}_c \;=\; \Pr\big[\, f(\, x \oplus \tau\,) = y_{\text{target}} \;\big|\; y(x) = c \,\big], \qquad c \neq y_{\text{target}}.$$

The conditional set $\{x : y(x) = y_{\text{target}}\}$ contributes no quantity to "flip to target": a target-class input is already at y_{target}, so the event "triggered prediction equals y_{target}" is true at baseline and does not represent attack success. Reporting $\mathrm{ASR}_{y_{\text{target}}} = 1$ would conflate baseline correctness with attack effect; reporting $\mathrm{ASR}_{y_{\text{target}}} = 0$ would conflate the absence of an attack with attack failure. We therefore record the cell as "N/A", a deliberate undefined value rather than missing data, and visualize it as gray (Figure 18, right panel, bottom row). For target-class samples, the analogous quantity of interest is the trigger's effect on output confidence, which is captured by the per-class ODS shown in the left panel.

Why ODS vanishes on the target class as scale grows. On target-class inputs, both p_{clean} and $p_{\text{triggered}}$ concentrate on the target label. Letting $p_{\text{clean}}(y_{\text{target}}) = 1 - \varepsilon_c$ and $p_{\text{triggered}}(y_{\text{target}}) = 1 - \varepsilon_t$ with $\varepsilon_c, \varepsilon_t \to 0$ as N grows, the Jensen–Shannon divergence between two near-point-mass distributions at the same atom is $O(\varepsilon_c^2 + \varepsilon_t^2)$, vanishing faster than the confidence gap itself. The empirical decay from $\mathrm{ODS}_{\text{class 1}} \approx 3.3 \times 10^{-3}$ at Tier-1 to 4.2×10^{-7} at Tier-7 is consistent with this regime: larger models become so confident on clean target-class inputs that no trigger perturbation can move the predictive distribution meaningfully.

Implication for the CRSC asymmetry. Two observations follow. First, essentially all of the ODS scaling is contributed by the non-target class , which is the only class the targeted backdoor can affect by design. Second, the per-class ASR on class 0 samples reaches 0.98 at Tier-4 and remains there, confirming that the backdoor flips the targeted class with very high reliability once embedded. The asymmetry between class 0 and class 1 is therefore a *structural property of the targeted backdoor threat model* (§3) and not an artifact of metric aggregation, masking, or sample-size imbalance. Untargeted backdoors, when added in future work, would distribute ODS and ASR more evenly across both classes.

Table 21: Per-class ODS and ASR on the rare token MLP setting. The target class (class 1) experiences essentially zero distributional shift; all of the trigger effect concentrates on the non-target class.

Tier	ODS class 0	ODS class 1 (target)	ASR class 0	ASR class 1
Tier-1	0.064	3.3×10^{-3}	0.106	n/a
Tier-2	0.103	2.3×10^{-3}	0.253	n/a
Tier-3	0.549	1.1×10^{-4}	0.965	n/a
Tier-4	0.592	3.6×10^{-5}	0.982	n/a
Tier-5	0.607	1.4×10^{-5}	0.983	n/a
Tier-6	0.626	3.8×10^{-6}	0.977	n/a
Tier-7	0.650	4.2×10^{-7}	0.983	n/a

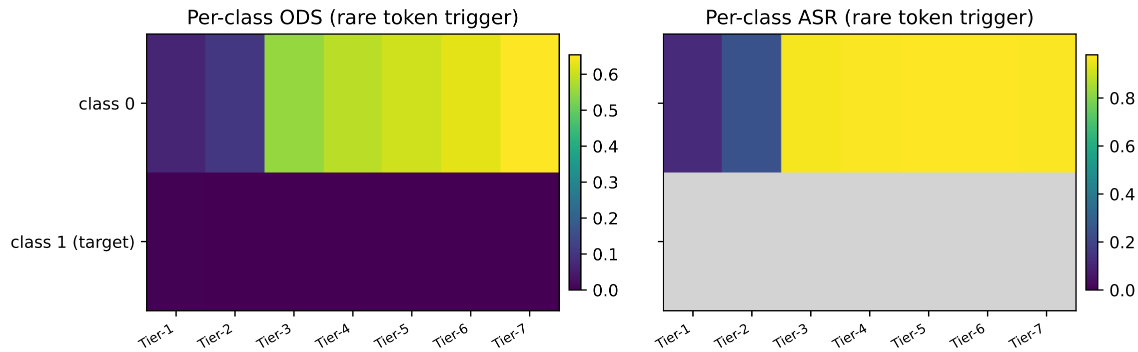

Figure 18: Per-class ODS (left) and per-class ASR (right) on the rare token MLP setting. Cell values are annotated; ODS values below 10^{-3} are shown in scientific notation. The asymmetry is structural rather than empirical noise: (i) the bottom row of the right panel is shown in gray and labelled "N/A" because ASR is *mathematically undefined* for the target class: ASR measures the fraction of non-target samples that get flipped to the target under trigger, a quantity whose denominator is empty when restricted to class 1 (target) inputs; (ii) the bottom row of the left panel is finite but vanishingly small ($\sim 10^{-7}$ at Tier-7) because large models predict class 1 with near-unit probability on clean target-class inputs, so the trigger-induced perturbation produces a negligible Jensen–Shannon divergence on those samples. Both gaps are predicted by the targeted-backdoor threat model (§3) and confirm that the trigger effect concentrates on the attack-relevant non-target class.

14 Discussion

RQ1 and RQ2. On MLPs the rare token trigger shows a positive scaling trend significant at the raw 0.05 level; on CIFAR-10 CNNs CRSC scales as $\sim N^{0.64}$ with $R^2 = 0.92$ and $p = 0.040$; on pretrained BERTs in the 4.4M to 29M window the trend is null or slightly negative. The same metric instantiates across the three settings without redefinition.

RQ3. The component ablation and the saturation contrast across triggers identify two regimes. When the trigger is weak (rare token MLP), BPS is the dominant scaling channel. When the trigger has fully embedded (syntactic, VPI topic on MLP; large CNNs), BPS saturates and ODS carries the scaling. The PCA loadings on PC1 are nearly equal, confirming that the three components track a common underlying signal.

RQ4. *The underlying functional form is saturating, not a strict power law*: AIC favors the logistic over the power law by $\Delta\text{AIC} = -26, -13, -10$ for the three MLP trigger families (Section 11). Reported β exponents are therefore local slopes of a saturating curve, not asymptotic scaling constants; they are retained alongside logistic parameters for comparability with the scaling-laws literature. The trend is robust to seed (Table 13), poison rate (Figure 10), ΔLSS normalization, and equal vs. PCA-derived weights (within 0.001). It is dissolved by both Bonferroni and Benjamini–Höchberg corrections across nine tests (Table 4, $p_{\text{BH}} \geq 0.077$ for all cells); given inter-component correlations ≥ 0.80 (Table 14), neither correction is well-calibrated here. FDR control with BH is the principled choice for future experiments at larger N where raw p-values will be smaller.

RQ5. Layer level patterns differ across architectures. MLPs concentrate trigger sensitivity in intermediate layers L2 to L4 for the rare token and spread it for structured triggers. Pretrained BERTs show very small per layer drops (≤ 0.05), suggesting subspace encoded pathways not surfaced by mean pooled cosine similarity. CNNs on CIFAR-10 show the largest sensitivity in the final convolutional layer for CNN-large, where the spatial patch is most fully integrated into class evidence.

Implications. The combined picture argues against the simple intuition that scale alone determines safety. CRSC works as a measurement tool in all three regimes; the sign and magnitude of the trend depend on architecture, pretraining, trigger family, and where in the AIC sense the model sits on the saturating curve. Linking scaling laws to cyber resilience is therefore a question of measuring rather than asserting how scale shapes fragility for a given pipeline.

Prediction-level contrast to Bowen et al. (2025). Bowen et al. [2025] predict monotone, unbounded ASR growth with scale. CRSC makes a stronger structural claim: the functional form is saturating (logistic), not unbounded. This prediction is directly testable: a follow-up with ≥ 5 CNN or BERT scales would distinguish whether the saturation in Figure 1 is a feature of our scale window or a genuine plateau. The diagnostic decomposition is also beyond Bowen's apparatus: when BPS saturates but ODS continues to grow, CRSC captures a signal that post-safety ASR cannot. The blended-trigger control (§9.2) is the direct falsification check.

Complementarity with detection metrics. CRSC is a *white-box continuous scalar* for resilience auditing, not a binary backdoor detector. NeuralCleanse [Wang et al., 2019], Spectral Signatures [Tran et al., 2018], Activation Clustering [Chen et al., 2019], and BackdoorPerturb [Huang et al., 2024] primarily ask whether a backdoor is present. CRSC instead measures its persistence, representational implication, and output-distribution consequence. A detector could first flag a compromised model; CRSC could then quantify its resilience across safety checkpoints and model sizes. A direct comparison on common checkpoints remains future work.

CRSC as downstream evaluation for persistence-aware attacks and defenses. Persistence-optimized attacks such as P-Trojan [Cui et al., 2026] align trigger gradients across safety fine-tuning steps to survive post-training. Embedding-drift defenses such as BEEAR [Zeng et al., 2024] regularize the safety stage to minimize trigger-induced representation change. Both operate on mechanisms that CRSC measures directly: BPS captures survival probability across checkpoints; $1 - \text{LSS}$ captures embedding drift; ODS captures the distributional consequence. CRSC could therefore serve as a standardized downstream evaluation metric for such methods, enabling direct comparison across attack/defense pairs on a common bounded scale without relying on ASR alone. We identify this as a high-value near-term extension; all CRSC components are available in the released pipeline with no retraining required.

Novelty attribution. In decreasing order of novelty: (i) the *cross-architecture instantiation* of a single formula (BPS, LSS, ODS) on MLP, transformer, and CNN without redefinition is the most novel element, requiring three pooling choices (no pool; mean-sequence-pool; GAP-pool) to be non-trivially equivalent in measurement intent; (ii) the *diagnostic decomposition* validated by the blended-trigger control is a genuinely new operational result; (iii) the *per-instance JS averaging* (Jensen-inequality argument in §4.3) is a methodological refinement over mean-distribution divergence; (iv) the *bounded composite* definition is incremental over multi-panel dashboards such as BackdoorLLM's ASR/clean-acc panels.

15 Limitations and Threats to Validity

(L1) MLP regime uses TF-IDF. The Setting A vectorizer discards word order; the bigram variant in Section 11 addresses this for the syntactic trigger and finds a different scaling sign, indicating partial bag of words artifact.

(L2) Pretrained transformer scale range. bert-tiny to bert-small spans 4–29 M total parameters, less than one order of magnitude, and $n = 3$ scale points means the Spearman rank test has minimum achievable $p = 0.167$ (one-sided), so no positive-trend claim can pass conventional significance in this setting by construction. The null or slightly negative direction ($\beta = -0.33$ for rare-token; $\beta = +0.04$ for syntactic, consistent with noise) is itself a substantive finding: it shows that scale-amplified fragility is architecture-dependent and is not a universal property of all neural classifiers. This is the operationally important point for practitioners who deploy pretrained transformers rather than from-scratch models. Adding `roberta-base` ($\sim 125\,\mathrm{M}$) as a fourth scale point would bring the window to two orders of magnitude and would, for the first time, allow a Pearson test with $n = 4$ and a Spearman test with minimum achievable $p = 0.083$, a meaningful threshold reduction. The CLS-token pooling sensitivity check (§8.1) is also required before any cross-architecture representational claim can be extended to the transformer family.

(L3) Safety procedure is clean SFT. All three settings use clean supervised fine-tuning as the safety stage. DPO and RLHF-style preference learning are not covered; these are the most important real-world extensions. The title and abstract use "SFT-Aligned" to reflect this scope exactly; the broader "post-trained" framing is reserved for a follow-up that includes DPO/RLHF.

(L4) Transformer LSS uses mean pooling. The per-layer drop is ≤ 0.05 on BERT, suggesting a subspace-encoded trigger pathway not surfaced by mean cosine similarity. The CLS-token cosine check (§8.1) is the next validation; CKA/SVCCA [Kornblith et al., 2019] would provide a more sensitive test at higher compute cost.

(L5) ODS at θ_T only. ODS is computed only at the safety endpoint, whereas BPS is averaged across checkpoints. Symmetric checkpoint averaging for ODS is supported by the code and a worthwhile extension; we report endpoint ODS for parsimony.

(L6) Construct validity of weights. PCA provides supporting evidence for the equal-weight choice (Section 11); a fully data driven scheme based on a downstream attack outcome regression is left to future work.

(L7) "Cyber resilience" framing is attacker-centric. A higher CRSC indicates a *more* deeply embedded trigger – *lower* defender-side resilience. Standard cyber resilience frameworks [Al Hidaifi et al., 2024, Tzavara and Vassiliadis, 2024] define resilience as the defender's capacity to resist and recover; CRSC measures the *inverse*: how resilient the backdoor is to removal. The naming reflects the attacker-centric perspective on compromise persistence. An alternative title – *Backdoor Persistence and Shift Index* – would avoid this ambiguity. Users should interpret a CRSC close to 1 as a system that *lacks* resilience against that trigger, not one that possesses it.

(L8) Architecture confounded with scale in Setting C. The four CNN tiers increase depth and width simultaneously, so parameter count covaries with architectural depth and with clean accuracy (which rises from approximately 0.33 to 0.54 across the tier range). The partial correlation analysis in Section 13.2 controls for clean accuracy on Setting C and finds partial $r = 0.99$ versus raw $r = 0.97$, providing weak evidence against a pure accuracy confound. However, a width-only or depth-only design with ≥ 8 controlled points and equalized training would be required to isolate the parameter-count effect from learnability threshold effects. The current evidence is associational; causal language ("scale *causes* increased persistence") should be avoided.

(L9) Vision trigger is BadNets only. The CIFAR-10 trigger is a single fixed 3×3 white patch, the earliest and least stealthy backdoor trigger family in the literature. Nguyen and Tran [2021] (WaNet), input-aware backdoors [Nguyen and Tran, 2020], and feature-space triggers represent more realistic modern attacks. The blended-trigger control (§12) addresses the stealthy end of the spectrum but does not constitute a modern attack trigger. Adding at least one WaNet-style trigger is a priority for the next version.

(L10) Statistical power. Setting B has 3 data points and Setting C has 4. These sample sizes cannot support tight bootstrap confidence intervals on β and the reported p-values should be interpreted cautiously.

For Setting B ($n = 3$), no test at conventional $\alpha = 0.05$ can achieve significance: the minimum one-sided p for Spearman rank correlation with $n = 3$ is 0.167. The results are therefore informative only about direction, not about magnitude or significance.

For Setting C ($n = 4$), the LOO analysis in Section 9.1 provides the appropriate sensitivity characterization: all four leave-one-out folds return a positive β (range 0.575–0.795 for CRSC, 0.474–0.658 for BPS, 1.370–1.675 for ODS), and the ODS component is perfectly monotone in N (Spearman $\rho = 1.000$, one-sided exact $p = 0.042$). The directional result is robust to the removal of any single point, even though p loses significance with $n = 3$ in each fold. The M_{eff}-adjusted Bonferroni correction (§10.4) reduces the effective number of tests from 9 to 3.87, placing the observed raw p-values (0.032–0.048) between the naive Bonferroni threshold (0.0056) and the lenient M_{eff} threshold (0.013). Both corrections remain unmet, but the true type-I error control is substantially less conservative than standard Bonferroni implies. Adding a fifth CNN scale (e.g., a ResNet-style block with $\sim 7\,\text{M}$ parameters) is the single highest-priority experiment to move the Setting C fit from 4 to 5 points and raise statistical power.

Table 22: Threats to validity and mitigations.

Type	Threat	Mitigation
Internal	MLP/BERT/CNN optimization may not converge to optimum	Multiple seeds and 95% CIs reported
External	Demonstration scale, not LLM scale	Formula-portable metric (MLP, BERT, CNN); LLM follow up with BackdoorLLM
Construct	CRSC weights are modeling choices	PCA derived weights, ablation, Tier-1 ratio variant
Conclusion	Power fit may overstate precision	Bootstrap CIs and per seed fits (Setting A); logistic vs power AIC comparison
Multiple testing	Bonferroni dissolves individual significance	FDR alternative discussed; correlations support reduced independence
Ecological	Synthetic MLP / small SST-2 / small CIFAR-10 subsets	Clean baseline ($\leq 2.3 \times 10^{-4}$), poison sweep, code released

16 Ethics and Responsible Research

All three settings produce only benign label flips under the trigger; no harmful natural language or image content is generated. SST-2 (GLUE) and CIFAR-10 are publicly distributed academic datasets. Pretrained BERT and CNN checkpoints / architectures are used under their respective open licenses. The reproducibility code is released for academic study of backdoor persistence with explicit dual use disclosure. Researchers extending to large language models should follow responsible disclosure norms documented in Carlini et al. [2024] and the benchmark protocols of Li et al. [2025].

Use of AI Writing Assistance and Data Provenance. The authors acknowledge the use of large language model (LLM) assistance for grammar refinement, structural drafting, and bibliography formatting in the preparation of this manuscript, in accordance with IEEE's Authorship and AI Policy. All scientific content, experimental design, data analysis, and conclusions are the sole responsibility of the human authors. After using this tool, the authors reviewed and edited all content and take full responsibility for the accuracy of every claim, reference, and quantitative result reported. No numerical results were produced by the LLM; all values were verified against the primary experimental outputs. Benchmark data are derived from the publicly available GLUE SST-2 sentiment corpus [Wang et al., 2018], the CIFAR-10 image classification benchmark [Krizhevsky, 2009], and a synthetic two-class sentiment corpus generated by the authors' own pipeline. The complete experimental pipeline, dependency-pinned environment, and all numerical results are released at `https://github.com/ufxa/Backdoor-Persistence-at-Scale/`, which constitutes the canonical code artifact for this paper.

17 Conclusion

We introduced the Cyber Resilience Scaling Coefficient (CRSC), a bounded unit-normalized composite scalar whose primary operational value is a *diagnostic decomposition*: CRSC + ODS jointly distinguish a persistent backdoor (high BPS, high CRSC) from a trigger that causes input-sensitivity without behavioral compromise (flat BPS, flat CRSC, scaling ODS). This distinction, demonstrated by the blended-trigger control in §9.2, is not achievable by any single-aspect metric. As a secondary contribution, CRSC shows a positive scaling trend on CIFAR-10 CNNs ($\beta = 0.64$, $R^2 = 0.92$, $p = 0.040$, $N = 4$; exploratory, pending replication; leave-one-out $\beta \in [0.575, 0.795]$ across folds reported in §9.1), and the underlying functional form is saturating (logistic fit preferred by AIC, $\Delta\text{AIC} = -26$ for the MLP rare-token regime). On pretrained BERTs the trend is null within the studied scale window (4–29 M total parameters), with mean-pooled LSS leaving open whether the null reflects a measurement artifact or a genuine architectural difference (see §8.1 and Limitation L4). We instantiated the framework across three architectural families under clean-SFT safety procedures (MLP, pretrained BERT, CNN) and characterized a sensitivity battery covering clean baseline, poison-rate sweep, per-seed exponents, PCA-derived weights, BH and Bonferroni corrections, ΔLSS, temperature scaling, ANOVA decomposition, T-checkpoint sweep, partial correlation, and per-class decomposition. The full pipeline is released at `https://github.com/ufxa/Backdoor-Persistence-at-Scale/`.

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